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The Hidden Power of Trade: Trade Networks as Mediators for Democracy on Human Rights Voting

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Abstract

How and why do states vary in their voting choices on human rights resolutions on the international stage? Although democracy is often seen as the primary determinant of foreign policy decisions on normative issues like human rights protection, autocratic states regularly offer yes votes on United Nations General Assembly (UNGA) human rights resolutions, sometimes more than established democracies do. We propose trade networks as a previously overlooked mediator of this relationship. Applying a recent regression-based network mediation model to nearly two decades of data, we estimate the indirect pathway from regime type to UNGA voting via trade network interdependence, alongside the direct pathway that does not run through the network. Our analysis reveals that the indirect effect of regime type via trade interdependence is substantial and tends to be positive, while direct effects of regime type tend to be negative. Combined, these can appear as null, slightly positive, or slightly negative average treatment effects of regime type on UNGA human rights voting, which helps explain prior mixed findings. These findings caution against treating regime-type effects on UNGA human-rights voting as evidence of normative alignment. What such votes do reveal is which conception of the international human rights regime a state is willing to back, and our results identify trade-network position as a distinct and countervailing influence on that choice. Our work identifies a previously neglected mechanism of trade network in international relations by adopting a novel statistical model.

Keywords: network mediation, causal mediation, trade networks, democracy, human rights

1. Introduction

Does regime type shape states' voting activity for human rights resolutions in international governmental organizations (IGOs)? Multilateral votes in bodies like the United Nations General Assembly (UNGA) can serve as signals of normative commitments and foreign policy positions while also structuring international pressure (Bailey et al., 2017; Kim & Russett, 1996). Thus, understanding their determinants is central to explanations of human-rights compliance, norm diffusion and external constraints on domestic politics. Likewise, the relationship between democratic rule and state repression remains a core concern in political science (Davenport & Armstrong, 2004). While democracies are often presumed more likely to support human rights measures (Davenport, 1999; Gerring et al., 2022), substantial variation remains across and within regime types (Beer & Mitchell, 2004; Cingranelli & Filippov, 2010), despite broad global improvements in rights over recent decades (Fariss, 2014).

In fact, the literature offers competing theoretical expectations and mixed empirical findings. Some accounts emphasize democratic norms and electoral accountability as drivers of support for human

rights (Hathaway, 2002; Hug & Lukács, 2014; Kim & Russett, 1996; Vreeland, 2008). Others stress strategic, economic or institutional constraints that can attenuate or reverse this effect (Moravcsik, 2000). These varying results suggest the presence of omitted mechanisms or heterogeneous causal pathways that condition the relationship between regime type and UNGA human rights voting.

We propose and evaluate a new mechanism in this setting, and posit that regime type affects UNGA human rights voting indirectly through international trade networks. Democratization can alter material incentives for signaling via shifts in where countries sit in the trade network and who they are most responsive towards in terms of market access and preferential trade obligations, thus changing the costs and benefits of voting yes on human rights resolutions. While prior studies have examined how economic incentives such as trade and foreign aid can influence UN voting behavior, the mediating role of trade networks in linking regime type to voting yes on human rights resolutions remains underexplored. To test this pathway, we employ a regression-based network mediation model, recently developed by Hayes *et al.* (2025), which allows the measured trade network itself to function as a mediator. Throughout, we treat UNGA voting as states' revealed positioning on a contested multilateral agenda rather than as a measure of underlying commitment to human rights *per se*.

Our paper makes three contributions. First, it specifies trade interdependence as a measurable channel through which regime type shapes multilateral human rights related UN voting. Second, it addresses a methodological gap by demonstrating a practical approach for estimating mediation (or indirect) effects when the mediator is a network, a problem that has constrained prior work because standard mediation techniques are ill-suited to relational data. Third, applying this approach to nearly two decades of UNGA voting, we find that democracy produces positive natural indirect effects via trade networks, but tends to produce negative natural direct effects. The opposing signs of these effects helps to reconcile the mixed findings in the existing literature.

2. Theory and Related Literature

Classical theories in the international relations literature posit that regime type systematically shapes state positions on human rights resolutions: democracies are expected to vote in favor both to credibly signal commitment to human rights norms (Hathaway, 2002, 2007; Hug & Lukács, 2014; Vreeland, 2008) and because domestic institutional constraints impose high costs on state repression, producing better on-the-ground records that naturally align with those resolutions (Donnelly, 1999; Landman, 2005; Simmons, 2009). Yet alternative research emphasizes strategic reticence among well-established, affluent democracies, which derive domestic legitimacy from entrenched human rights protections and therefore face limited marginal benefit—and nontrivial sovereignty costs—from endorsing additional international commitments. In contrast, newly democratized and post-communist states might have stronger incentives to vote for such resolutions to signal liberalization and accrue external credibility (Hafner-Burton *et al.*, 2015; Moravcsik, 2000).

The relationship between regime type and states' voting on human rights resolutions is theoretically ambiguous and empirically mixed: while some theories predict that higher levels of democratization increase the propensity to support human rights resolutions, others allow for countervailing dynamics that can attenuate or reverse such an effect. Empirically, findings vary. Several studies report a positive association (democracies are more likely to vote “yes” on human-rights resolutions, align in Western or liberal blocs, and ratify rights-relevant treaties more frequently; Boockmann & Dreher, 2011; Hathaway, 2007; Voeten, 2013). Other work finds null or negative relationships between democracy and human rights voting: Hug and Lukács (2014) find null or limited impacts of democracy on UN Human Rights Council voting, while Primiano and Xiang (2016) documents cases where autocratic states with poor rights records vote in favor of UNGA human rights resolutions.

These competing accounts share an observable implication: they are adjudicated by how states

position themselves when the multilateral human rights agenda is brought up to vote. We take this positioning seriously as an object of study in its own right. UNGA human rights resolutions are not simple or neutral referenda on whether rights matter, but are often contested instruments through which states advance rival conceptions of what human rights require (e.g., liberal and civil political framings versus sovereignty-based and developmental framings; Donnelly, 1999; Moravcsik, 2000). A state's pattern of voting thus reveals which conception of the international rights regime it is willing to bolster support behind and at what diplomatic cost. This matters substantively for at least two reasons. First, these votes are constitutive: they determine which texts are adopted and which framings accumulate increased legitimacy. Second, they are consequential: positions taken in UN human rights bodies carry reputational and material weight, as documented most clearly for country-specific condemnations, where resolutions have been linked to international shaming dynamics and multilateral resource flows (Hafner-Burton, 2008; Lebovic & Voeten, 2006, 2009).

What explains these discrepant findings on regime type and human rights voting? We propose that an under-examined mechanism—trade networks—helps reconcile these mixed results: regime type shapes patterns of bilateral and multilateral trade, and different trading networks socialize states into distinct expectations about how to position themselves on contested multilateral items. A country's trade network ties alter the marginal costs and benefits of voting yes on human rights resolutions by shifting whose markets, preferences, and reputational pressures matter most. These shifts ultimately affect whether democracies (or autocratic regimes) find it more advantageous, on balance, to vote “yes”.

More specifically, democratization is associated with changes in a state's embeddedness in a global trading system, who it trades with, how concentrated those ties are, and how those partners are connected (or not) to one another, rather than with a single scalar such as trade volume. Two mechanisms are compatible with this pathway, and we remain agnostic about their relative weights. The first centers around coalition alignment (and to some extent, homophily). As political institutions liberalize, governments can face stronger incentives to sustain access to large, rules-based markets and manage reputations among partners whose domestic politics might make human rights politics more salient. Trade ties can therefore cluster among states with more compatible institutional and policy commitments, generating voting alignment on contested resolutions through shared incentives and reputational politics in international institutions (Mansfield & Pevehouse, 2006; Russett & Oneal, 2001). The second mechanism operates via diffusion and socialization through economic networks: repeated economic exchange can transmit information, expectations, and coordination practices through partners and partners-of-partners, so that positions in the trade network capture more than simple dyadic dependence (Hafner-Burton et al., 2009). At the same time, deep integration need not be automatically followed by domestic rights convergence. Cases such as China and Vietnam illustrate that extensive trade with democracies can coexist with persistent divergence in domestic rights politics, which cautions against interpreting our mediation solely as “trade causes liberal norms at home.” Autocracies differ in the political meaning of trade embeddedness: because autocratic governments frequently prioritize regime security, selective rents, and bilateral bargains over rights-based reputational politics, their trade neighborhoods need not generate the same incentives for convergence on human rights votes, even when they are highly integrated. Thus, the mapping from network position to UNGA alignment can be weaker or structured by different strategic logics than among democracies (Davis & Pratt, 2021; Maoz & Russett, 1993).

This coalition-alignment logic does not require that a yes vote express sincere endorsement of the right at issue. Because much of the recurring UNGA human rights agenda pairs rights language with sovereignty-forward or developmental framings, states divide on these items along coalitional lines as much as along human rights practice lines (Boockmann & Dreher, 2011). We argue that trade-network position can shape which side of that division a state finds it costly or advantageous to join, and therefore which conception of the rights regime accumulates support and legitimacy.

To some degree, this logic resonates with classic Kantian peace theory, which posits that commerce serves as a major channel connecting democracy, peaceful relations, and human rights protection (see Mansfield & Pevehouse, 2000; Pevehouse & Russett, 2006; Russett & Oneal, 2001, among others). Regime type significantly affects a country's cooperation and conflict behaviors, as it shapes national interests and norms (Hagan, 1994; Maoz & Russett, 1993; Russett *et al.*, 1995). It further influences countries' economic cooperation, such as the scale and nature of trade policies: democracies are more likely to trade with each other, set lower barriers, and foster trade openness in developing countries (Chen *et al.*, 2023; Mansfield *et al.*, 2000, 2017; Milner & Kubota, 2005). Trade relations also reflect geopolitical interests, as countries tend to exchange goods with their political and military allies (Gowa & Mansfield, 1993; Long & Leeds, 2006). Countries often link issues of security and the economy to determine their bilateral and multilateral trade policies, making trade networks endogenous to regime type, security, and military considerations (Davis & Pratt, 2021).

While scholars have explored the relationship between regime type and various state behaviors, such as trade liberalization (Milner & Kubota, 2005) or civilian treatment in war (Appel & Prorok, 2019), either potentially through extrinsic pressure or intrinsic norm sharing and socialization (Greenhill, 2010), we extend this literature by explicitly theorizing and empirically testing how trade networks mediate the effect of regime type on human rights voting. Specifically, we argue that as countries transition along the democracy-autocracy spectrum, their evolving trade partnerships and embeddedness in international trade shape incentives to align with key trading partners' positions on UNGA human rights resolutions.

A different causal ordering is also plausible: deeper economic integration with democracies could raise the likelihood of democratization or shift domestic human-rights practices, which would then be reflected in UNGA voting (Mansfield & Pevehouse, 2006). We do not deny that this trade $\rightarrow$ regime (or rights) $\rightarrow$ vote pathway may operate in some contexts and horizons. Our objective, however, is to articulate and estimate a complementary ordering in which regime type structures trade-network embeddedness, which in turn mediates incentives for human rights positioning in the UN. Theoretically, privileging the latter is not arbitrary: a large international political economy literature treats domestic institutions and coalitions as central to trade openness and partner choice, so trade patterns are plausibly shaped by regime politics rather than serving only as a precursor to regime change. For example, Milner and Kubota (2005) argue that democratization reallocates power away from protectionist elites toward broader constituencies that benefit from openness, generating regime-linked trade liberalization in developing countries. Mansfield *et al.* (2000) show that democracies trade disproportionately with other democracies, underscoring how regime type structures partner choice and clustering. By contrast, the reverse story emphasizes domestic transformation through external economic exposure over time. Our focus is on downstream foreign policy implications of contemporaneous regime type for where states sit economically in the system and how that position relates to multilateral human-rights diplomacy. Empirically, we align this substantive ordering with a measurement scheme that sequences treatment, mediator, and outcome to reduce post-treatment bias when interpreting mediation along the path we foreground (see Section 3.2).

We particularly expect the mediating effect of trade relations to be pronounced among exporting countries, as they are more susceptible to economic incentives to adjust or signal foreign policy alignment. Export dependence creates concentrated economic stakes, and, in combination with organized lobbying to preserve market access (see Kim & Milner, 2021; Kim & Osgood, 2019, among others), exposes firms and states to buyer-driven human rights standards. Loss of large markets becomes a salient cost, especially for export-reliant countries. Consequently, countries democratizing from low levels of civil liberties are especially likely to adjust UN votes toward more frequent yes votes on human rights resolutions to avoid losing access to major democratic markets (Bernard *et al.*, 2009; Greenhill *et al.*, 2009; Hafner-Burton, 2005; Yasar, 2013).

Our theoretical framework aligns with previous research showing that international economic

competition and preferential trade agreements shape domestic policy and the adoption of human rights standards (Hafner–Burton, 2005; Simmons & Elkins, 2004). It also accords with findings that UN votes often reflect materialistic interests and strategic policy alignments, such as vote buying and aid in exchange for support (Brazys & Panke, 2017; Carter & Stone, 2015; Dreher & Sturm, 2012; Gallop & Minhas, 2021; Hagan, 1989). That literature treats UN roll calls as strategically chosen public positions rather than sincere expressions of principle. We adopt the same conception, while emphasizing that strategically chosen positions are still the material from which the multilateral rights regime is assembled. Building on these studies, we propose that regime type influences human rights voting in international organizations in part through trade interdependencies. This indirect pathway is distinct from any direct effect of democratization (or autocratization) arising from expanded civil liberties and rights or adoption of rights norms.

This distinction matters for what our hypotheses claim. A state's conduct on human rights has two faces that need not move together: what it does at home, through legislation, enforcement and treaty compliance, and what it says in multilateral forums, through the positions it takes when resolutions are put to a vote. Our theory speaks to the second. What we observe and model is voting behavior on UNGA human rights resolutions: whether a state votes yes, abstains, or votes no on resolutions classified under the human rights agenda. We read an affirmative vote as public endorsement of the resolution at hand rather than as evidence of rights protection at home. We treat an affirmative vote as public endorsement of the resolution at hand rather than as evidence of rights protection at home. Section 3.1 details how we construct this measure and what it does and does not capture.

From this theoretical framework, we derive two related testable hypotheses about human rights voting behavior:

H1 (Indirect effect of democracy via trade networks): Trade networks mediate the effect of a state's level of democracy on its human rights voting behavior.

H2 (Direct effect of democracy): A state's level of democracy impacts its human rights voting behavior, even when controlling for mediating mechanisms such as trade networks.

One reason this trade-network mechanism may have been understudied relates to the inherent challenge of modeling and estimating the mediating role of a network-valued random variable. Existing approaches often reduce inter-state relations to summary measures (e.g., counts of allies or trade partners Boehmer et al., 2004; Onderco, 2014; Pevehouse & Russett, 2006) or estimate dyadic outcomes with network-related controls (Hafner–Burton & Montgomery, 2006; Maoz, 2009). These strategies have two limits. First, they induce overcontrol bias when conditioning on the mediator (Cinelli et al., 2024). Second, they treat network structure as exogenous rather than as a pathway that is itself influenced by the treatment.

Moreover, relying on simple summary statistics to capture inter-state relations is problematic, as it overlooks crucial relational information. Measures such as membership in international governmental organizations (IGOs), military alliances, or trade volume have been widely used as proxy networks (Dorussen & Ward, 2008, 2010; Pevehouse & Russett, 2006). However, counting joint memberships or connected edges within these networks fails to capture the underlying relational dynamics, as they fail to capture embeddedness. By way of example, consider the Gulf War. The number of Kuwait's alliance ties alone cannot explain why states such as South Korea or Bangladesh deployed troops after Iraq's invasion. What matters is Kuwait's position in the broader alliance structure—who its partners are (e.g., Saudi Arabia, U.S.), who the partners' partners are, how they are connected to one another (e.g., Bangladesh's protection of its ties to Saudi Arabia; South Korea's alliance with the U.S.), and how influence propagates through partners-of-partners. Aggregating network measures can also introduce bias in estimated values, ultimately leading to misleading inferences (Kim et al., 2019; Lo et al., 2025; Relión et al., 2019). Our network approach captures these higher-order dependencies

by using low-dimensional embeddings of countries' positions in the trade network, which encode structural roles and neighborhood context beyond first-order trade ties.

To address concerns around overcontrol bias and incorporate network structure, we employ a recently-developed statistical method that enables the evaluation of causal mediation mechanisms when the mediator is measured as a network. This network mediation model, proposed by Hayes *et al.* (2025), leverages spectral embeddings (Sussman *et al.*, 2014) to extract low-dimensional representations of nodes, which are assumed to be noisy proxies for latent relational structure. The model treats latent homophily within a network as a mediating pathway linking a node-level treatment to nodal outcomes, providing a principled approach to estimating indirect effects operating through networks while preserving full relational information. To our knowledge, no alternative approach exists that simultaneously a) operates within the potential outcomes framework and b) incorporates the network mediator directly into the model.

This study contributes to the literature on the impacts and limits of regime type on human rights voting by proposing a new theoretical mechanism: trade interdependence. Specifically, we extend the literature examining extrinsic, interest-based foreign policy decisions and explore the downstream consequences of trade interdependence on UNGA voting. Methodologically, we introduce and illustrate from the statistics literature a novel causal inference-oriented modeling and estimation approach for social science research questions where networks act as mediators. Together, the theoretical argument and the network mediation framework define what we estimate: how regime type relates to UNGA human rights voting, and how much of that relationship operates through where states sit in the trade network space rather than through a separate, direct pathway. The next section describes the data and methods employed, including a more detailed explanation of the network mediation model.

3. Data & Methods

3.1 Data

To evaluate how regime type influences human rights voting via trade networks, we assemble measures of regime characteristics, bilateral trade ties, and relevant confounders, most notably contemporaneous economic capacity and prior conflict involvement, which are included as controls. To ensure temporal ordering and reduce simultaneity bias, control variables are measured prior to the main explanatory variable (regime type), which is measured prior to the mediator (trade network), which is measured prior to outcome (UNGA human rights voting). For example, regime type and trade-network attributes recorded in 2014 and 2015, respectively, are used to predict human-rights voting observed in 2016.

Independent variable: Regime Type We use data from the Varieties of Democracy (V-Dem) project (Coppedge *et al.*, 2025), which distinguishes between five high-level principles of democracy: electoral, liberal, participatory, deliberative and egalitarian. Our main independent variable is the electoral democracy index (or polyarchy score), obtained from the *vdem* R package (Maerz *et al.*, 2019, November 29/2025). The polyarchy index captures key elements of electoral democracy, including freedom of association, freedom of expression, suffrage, free and fair elections, and the selection of the chief executive and legislature through elections. This aligns with the theoretical focus on electoral constraints in existing research on regime types (Bogati *et al.*, 2025; Cottiero & Haggard, 2023; Teorell & Lindberg, 2019).

Outcome variable: UNGA Human Rights Voting Our outcome is states' voting behavior on United Nations General Assembly (UNGA) resolutions belonging to the human rights issue category. We do not interpret affirmative votes as a direct measure of a country's underlying commitment to human rights. Indeed, many UNGA human rights resolutions are recurring and politically

contested: governments may oppose or abstain because they take issue with the resolution's framing or sovereignty implications while others might vote "yes" with strategic reasons in mind (Boockmann & Dreher, 2011; Primiano & Xiang, 2016). Accordingly, we treat higher values of the outcome as greater support for UNGA human rights resolutions (equivalently, human rights related UN voting). Although we don't equate affirmative votes with sincere commitment to human rights, UN human rights votes are public acts that governments, international organizations, and advocacy networks treat as positions in the multilateral human rights agenda, with reputational and material consequences (Lebovic & Voeten, 2006, 2009). We therefore see this outcome as capturing meaningful foreign policy behavior in UN voting.

This measurement choice is standard when the research question concerns behavior on a specific UN agenda rather than latent foreign policy ideology. Ideal point models of UNGA voting, such as in Bailey et al. (2017) and Voeten (2013), recover coalitional preferences by allowing resolutions to differ in discrimination and difficulty. They are also typically interpreted as alignment with the U.S. led international order rather than as support for human rights. By contrast, a large literature analyzes votes on human rights resolutions themselves, both in the UNGA and related bodies, as the substantive outcome of interest when studying rights politics in multilateral spaces (Boockmann & Dreher, 2011; Hug & Lukács, 2014; Prasad & Nooruddin, 2024; Primiano & Xiang, 2016). Because our theory concerns how trade network position mediates regime type's effects on states' public positions on the UNGA human rights agenda, aggregating roll call choices on human rights resolutions is appropriate for our estimand. We characterize the composition of this agenda, including prevalence of recurring and contested items, in Appendix Table 4, and we assess sensitivity to particularly politicized resolution categories in our robustness checks (see Appendix Tables 6–7). We use the `unvotes` R package (Robinson, 2016), which provides historic voting data from the UNGA, to obtain votes for each country in each roll call, as well as descriptions of topic classifications for each vote. Specifically, we merge the `un_votes`, `un_roll_calls`, and `un_roll_call_issues` datasets to include roll call vote IDs, descriptions, and results from all member countries. Our primary outcome variable, *vote_n*, is coded as 1 if a country voted "yes," 0.5 if it abstained, and 0 if it voted "no." As a robustness check, we also construct a binary variable, *vote_yes*, where abstentions are coded as 0, along with "no" votes. Example resolutions are presented in Table 1. As the merged UNGA votes data ends in 2019, our analysis ends with the year 2019.

We select our analysis window based on data availability and coverage consistency, yielding models for UN voting years 2001–2019 (see Appendix Table 3 for a full summary of utilized data and years). These years capture critical regime shifts in U.S. presidential administrations, which significantly affect global political and economic alignment (Carter & Stone, 2015; Mosler & Potrafke, 2020). Specifically, 2003 reflects the Bush administration's "War on Terror" era, while the 2010s capture the period of Obama's administration, where multilateralism and cooperation were far more emphasized. We expect that a pivotal global realignment affecting human rights votes took place in the year 2015, following the geopolitical shocks of 2014: Russia's annexation of Crimea, the rise of ISIS, and pro-democracy protests in Hong Kong. Finally, 2017 and 2019 under Donald Trump represent a time of "America First" foreign policy, along with the aftermath of Brexit and the escalation of the U.S.–China trade war. Our temporal measurement strategy follows a fixed causal sequence. If the treatment is measured at time t , pre-treatment covariates are measured at $t - 1$, the mediator at $t + 1$, and the outcome at $t + 2$. We use this timing structure in the main specification to align estimation with the hypothesized causal pathway and reduce risks of post-treatment bias.

Mediator: Trade networks We operationalize the mediator using bilateral trade flows from the World Trade Flow Characterization (WTFC) dataset (Emlinger & Piton, 2014), measured as exports from country i (exporter) to country j (importer). The dataset yields a weighted network whose edge weights are log total dollar value trade flows. We incorporate multiple cross-sectional trade

networks (2000–2018; again, one year prior to measurement of the outcome variable) to capture temporal variation in trade relations. Figure 1 presents an example year’s trade network, where nodes represent countries and edges encode trade flows between countries.

Control variables Because prior economic conditions, exposure to conflict and instability, and preexisting domestic human rights practices may jointly affect regime type, trade-network structure, and states’ UNGA voting on human rights resolutions, we assemble a set of potential confounders measured in the year prior to the independent variable, as well.

Economic capacity is proxied by log GDP from the World Development Indicators and Other World Bank Data (Arel-Bundock, 2010). We similarly measure population size with logged population from the WDI data. These factors simultaneously impact democratization paths and trade dependencies (a diagram of our causal pathways is given in Figure 2), but also can influence a country’s stance on multilateral human rights agenda items (De Mesquita *et al.*, 2005). We also include membership in major regional and international organizations (World Trade Organization, European Union, OIC) to account for institutional and normative pressures that shape voting behavior (Cao *et al.*, 2013; Hafner-Burton, 2005; Hug & Lukács, 2014).

Prior conflict history has been tied to regime type (Hegre *et al.*, 2001) and state human rights record (the presence of political violence can exacerbate states’ positions on human rights foreign policy due to a fear of reciprocal condemnation from foreign countries; see, for instance, Prasad & Nooruddin, 2024). Given the various forms it can take, we consider within- and across- state conflicts together when selecting measures. We use measures of inter- and intra-state conflict (lagged, continuous) drawn from a collective project between the Department of Peace and Conflict Research at Uppsala University and the Centre for the Study of Civil War at the Peace Research Institute Oslo, referred to as the UCDP-PRIO Armed Conflict Dataset (Gleditsch *et al.*, 2002).

Finally, we control for prior state respect for civil liberties using V-Dem’s human rights index to account for preexisting variation in domestic human-rights practices (Coppedge *et al.*, 2025). In Section 3.2, we detail how these data are utilized in the our network mediation regression model and fully describe our research design.

3.2 Methods

We adopt a recently developed approach for network-based mediation analysis (Hayes *et al.*, 2025) to evaluate whether regime type affects human rights voting either directly or indirectly via trade-network interdependence. Hayes *et al.*, 2025 model the observed network as a random dot product graph (RDPG; Athreya *et al.*, 2018), a model in which each node $i \in [n] = \{1, 2, \dots, n\}$ has an associated *latent position* vector $X_i \in \mathbb{R}^k$ that encodes the “structural role” of the node. For example, in the stochastic blockmodel (a special case of the RDPG), these latent positions can be thought of as encoding each node’s community memberships. In our setting, we have a network whose nodes correspond to countries, and edges encode trade flows between countries. Thus, we may conceive of these latent positions as encoding how countries sit vis à vis one another in a latent “trade space”, which is noisily observed via (log) trade volume. The estimator in Hayes *et al.*, 2025 is a semi-parametric network regression that treats node latent positions as causal mediators, thereby providing a principled basis for estimating indirect effects that operate through relational structure.

Compared with alternative network mediation procedures, the method permits causal effect estimation while explicitly accounting for homophily and network dependence under weaker assumptions (Chang & Paul, 2024; Li *et al.*, 2019; Liu *et al.*, 2021). Because the model targets an unobserved latent mediator (i.e., the latent positions $X_1, X_2, \dots, X_n$), it requires only a proxy network that captures the general structure of the true relations rather than a perfectly measured mediator. In our setting, trade flows between countries serve as this measured mediator. The Hayes *et al.*, 2025

estimator accommodates weighted edges, measurement noise in edge observations, and heteroskedasticity in network errors. In particular, we need only assume that the observed network is a noisy version of the latent structure, in which errors in edge measurements are (conditionally) independent with suitably well-behaved tails. Our first assumption concerns this network measurement error structure. For simplicity, we state this and subsequent assumptions for the case of a symmetric adjacency matrix, but this can be extended straightforwardly to the case of directed networks, such as in our case of country trade networks.

Assumption 1 (Sub-gamma Network). Let $A \in \mathbb{R}^{n \times n}$ be a random symmetric matrix, such as the adjacency matrix of an undirected graph. The conditional expectation of A conditional on $X \in \mathbb{R}^{n \times k}$ is given by $\mathbb{E}[A \mid X] = XX^T$, in which X has independent and identically distributed rows $X_1, X_2, \dots, X_n$. The matrix XX^T has rank k and is positive semi-definite with eigenvalues $\lambda_1 \geq \lambda_2 \geq \dots \geq \lambda_k > 0 = \lambda_{k+1} = \dots = \lambda_n$. Conditional on X , the upper-triangular elements of $A - XX^T$ are independent (v_n, b_n) -sub-gamma random variables (Boucheron et al., 2013).

We briefly introduce some notation to present the model, assumptions and quantities of interest, along with our estimation approach. We follow a potential outcomes framework throughout (Neyman, 1923; Rubin, 2005). We let $T \in \{0, 1\}^n$ be a vector of observed node-level binary treatment indicators¹ and $Y \in \mathbb{R}^n$ be a vector of observed node-level outcomes. We collect the node-level latent positions $X_1, X_2, \dots, X_n$ in the rows of $X \in \mathbb{R}^{n \times k}$, so that the row vector $X_i \in \mathbb{R}^{1 \times k}$ denotes the mediator for node $i \in [n]$. Similarly, $C \in \mathbb{R}^{n \times p}$ is a matrix of observed confounders, with i -th row $C_i \in \mathbb{R}^{1 \times p}$ denoting confounders for node i . $A_{ij} \in \mathbb{R}$ denotes the weight of the edge joining nodes i and node j .

In this set up, we have three quantities of interest: the average treatment effect (ATE), the natural direct effect (NDE) and the natural indirect effect (NIE). We interpret these estimands, as defined below, in line with VanderWeele (2015). Let $Y_i(t)$ be the counterfactual value of the outcome measured for the i -th node when node treatment T_i is set to t . Similarly, let $Y_i(t, x)$ be the counterfactual value of the i -th outcome when T_i is set to t and X_i is set to x , and let $X_i(t)$ be the counterfactual value of the mediator X_i when T_i is set to t .

Definition 1 (Average Treatment Effect). The *average treatment effect* Ψ_{ATE} describes how much the outcome Y_i would change on average if the treatment T_i were changed from $T_i = t$ to $T_i = t^*$:

$$\Psi_{\text{ATE}}(t, t^*) = \mathbb{E}[Y_i(t) - Y_i(t^*)].$$

Definition 2 (Natural Direct/Indirect Effect). The *natural direct effect* (NDE) describes how much the outcome Y_i would change if the exposure T_i were set at level $T_i = t^*$ versus $T_i = t$ but for each individual the mediator X_i were kept at the level it would have taken for that individual, had T_i been set to t^* :

$$\Psi_{\text{NDE}}(t, t^*) = \mathbb{E}[Y_i(t, X_i(t^*)) - Y_i(t^*, X_i(t^*))].$$

The *natural indirect effect* (NIE) describes how much the outcome Y_i would change on average if the exposure were fixed at level $T_i = t^*$ but the mediator X_i were changed from the level it would take under $T_i = t$ to the level it would take under $T_i = t^*$:

$$\Psi_{\text{NIE}}(t, t^*) = \mathbb{E}[Y_i(t, X_i(t)) - Y_i(t, X_i(t^*))].$$

In our setting, the average treatment effect, which is the sum of NIE and NDE, corresponds to the effect of increased democracy on a state's probability of voting "yes" on UNGA human-rights resolutions. The NIE captures the component of the democratic treatment that operates through

¹For concreteness and illustration purposes we present a binary T here, though note the democracy score in our setting ranges between 0 and 1 and notation follows accordingly.

changes in trade interdependence. Specifically, the effect of increased democracy on UNGA human rights voting transmitted via alterations in the structure and strength of bilateral trade ties. This quantity corresponds directly to the theoretical mechanism tested in hypothesis H1. The NDE captures the component of the democratic treatment that affects UNGA human rights voting independently of trade network interdependence (that is, the effect of increased democracy holding the trade network fixed). This quantity corresponds to the theoretical mechanism tested in H2. We present an illustration of our theoretical causal pathways in Figure 2.

Three standard assumptions are required for identification and estimation of our causal estimands in Definitions 1 and 2, which we collect under Assumption 2:

Assumption 2 (Non-parametric Identification of Natural Direct and Indirect Effects). The random variables $(Y_i, Y_i(t, x), X_{i\cdot}, X_{i\cdot}(t), C_{i\cdot}, T_i)$ are independent over $i \in [n]$ and obey the following three properties.

1. **Consistency:** if $T_i = t$, then $X_{i\cdot}(t) = X_{i\cdot}$ with probability 1, and if $T_i = t$ and $X_{i\cdot} = x$, then $Y_i(t, x) = Y_i$ with probability 1
2. **Positivity:** $\mathbb{P}(x | T_i, C_{i\cdot}) > 0$ for each $x \in \text{supp}(X_{i\cdot})$ and $\mathbb{P}(t | C_{i\cdot}) > 0$ for each $t \in \text{supp}(T_i)$
3. **Sequential ignorability:** $\{Y_i(t^*, x), X_{i\cdot}(t)\} \perp\!\!\!\perp T_i | C_{i\cdot}$ and $\{Y_i(t^*, x)\} \perp\!\!\!\perp X_{i\cdot} | T_i = t, C_{i\cdot}$.

Assumption 2 invokes standard identification conditions for causal mediation. Consistency requires well-defined interventions and that the observed trade ties and voting outcomes correspond to the potential mediator and outcome values under those interventions: observed values equal their respective potential outcomes under the realized treatment and mediator. Positivity speaks to the required sufficient support in the data across treatment and mediator values so that, conditional on covariates, all relevant combinations of treatment and mediator occur with positive probability. We also adopt the mediation analogue of selection-on-observables: conditional on pre-treatment covariates C , the treatment is as-if random to potential outcomes, and conditional on treatment and C , the mediator is as good as randomly assigned (Imai *et al.*, 2010). Formally, $T \perp\!\!\!\perp \{X(t), Y(t, x)\} | C$ and $X \perp\!\!\!\perp \{Y(t, x)\} | T, C$ for the relevant (t, x) which together permit identification of the natural direct and indirect effects under the model.

Central to our identification strategy is satisfying the sequential ignorability assumption. To render this assumption more plausible, we control for an extensive set of pre-treatment confounders that plausibly jointly influence regime type, trade-network structure and voting behavior (see Control Variables in Section 3.1 above); we also conduct a series of robustness tests in the Appendix to probe the sensitivity of our findings.

Our final assumption relates to specifying the outcome and mediator models.

Assumption 3 (Linear Conditional Expectations). The outcome regression functional is linear in $T_i, C_{i\cdot}$, and $X_{i\cdot}$; and the mediator regression functional is linear in T_i and $C_{i\cdot}$:

$$\mathbb{E}[Y_i | T_i, C_{i\cdot}, X_{i\cdot}] = \beta_0 + T_i\beta_t + C_{i\cdot}\beta_c + X_{i\cdot}\beta_x, \quad (\text{outcome model})$$

$$\mathbb{E}[X_{i\cdot} | T_i, C_{i\cdot}] = \theta_0 + T_i\theta_t + C_{i\cdot}\theta_c \quad (\text{mediator model})$$

We can also rewrite Assumption 3 in simpler notation by collecting the node-level covariates into $W = [1 \ T \ C] \in \mathbb{R}^{n \times (p+2)}$, where p is the number of confounder variables.

$$\mathbb{E}[Y_i | W_{i\cdot}, X_{i\cdot}] = W_{i\cdot}\beta_W + X_{i\cdot}\beta_X, \quad (\text{outcome model})$$

$$\mathbb{E}[X_{i\cdot} | W_{i\cdot}] = W_{i\cdot}\Theta \quad (\text{mediator model})$$

To fit the above models, we need access to the latent (trade) positions. Of course, these latent positions are unobserved. Thus, in order to fit these two models, we must estimate the latent positions

from the observed network A (our observed trade flow network). To do this, we apply the adjacency spectral embedding (ASE; Sussman et al., 2012), which yields an estimate of X , which we denote as $\hat{X}$. Under standard modeling assumptions, $\hat{X}$ converges to the true X uniformly over the rows of X , up to orthogonal non-identifiability (Levin et al., 2022; Lyzinski et al., 2014). We can plug these estimates in for X in the outcome and mediator models above and obtain least-squares point and variance estimators for the regression coefficients (for details, see Definitions 5 and 6 in the Appendix).

More formally, the ASE is defined as follows.

Definition 3 (Adjacency Spectral Embedding (Sussman et al., 2014)). Given a network with adjacency matrix A , the k -dimensional *adjacency spectral embedding* (ASE) of A is defined as

$$\hat{X} = \hat{U}\hat{S}^{1/2} \in \mathbb{R}^{n \times k},$$

where $\hat{U}\hat{S}\hat{V}^T$ is the rank- k truncated singular value decomposition of A . That is, $\hat{S} \in \mathbb{R}^{k \times k}$ is diagonal, with entries given by the k leading singular values of A , and $\hat{U}, \hat{V} \in \mathbb{R}^{n \times k}$ have the corresponding k orthogonal singular vectors as their columns.

Of course, in practice, we do not know the true dimension of the latent positions, and we must determine k from the data. We select k using the cross-validated eigenvalue procedure of Chen et al. (2021), using the `eigcv` function from the `gdim` package in R. This is roughly analogous to fitting on a train set and predicting on a test set. We further tested the sensitivity of these chosen values for k by repeating the bootstrap procedure multiple times with varying random seeds. Our resulting choices of k for the 2000–2018 trade networks ranged from $k = 10$ to $k = 12$ (see Appendix Figure 10).

The model we specify includes several parameters to be estimated. First, there are the network regression coefficients β and Θ from the outcome and mediator models (Assumption 3). Once we estimate the coefficients β and Θ via linear regressions, we can plug them into the outcome model and mediator model equations to obtain estimates of the natural direct and indirect effects, respectively. We estimate these causal quantities using the following empirical models and the observed data described above in Section 3.1:

$$\text{Vote}_i = \beta_0 + \text{Democracy}_i \cdot \beta_t + \text{Controls}_i \cdot \beta_c + \text{Trade position}_i \cdot \beta_x + \varepsilon_i, \quad (\text{outcome model})$$

$$\text{Trade position}_i = \theta_0 + \text{Democracy}_i \cdot \theta_t + \text{Controls}_i \cdot \theta_c + \eta_i \quad (\text{mediator model})$$

The outcome variable, Vote_i , represents each country's roll-call votes on human rights resolutions in the UN General Assembly (UNGA). The treatment variable, Democracy_i , indicates country i 's V-Dem polyarchy score (again, measured two years prior to the year of the outcome). Control variables (measured prior to treatment) include logged GDP, logged population, current human rights practices, the presence or absence of inter- and intra-state conflict, and one-year lags of these conflict variables. We also include membership status in the WTO, EU, and OIC as additional controls and account for unique resolution identifiers. Trade position $_i$ refers to the spectral embedding of country i within the trade network (the mediator, measured after treatment and before outcome variables), and ε_i and η_i are random errors.

4. Results

We estimate separate network mediation models of UNGA human rights voting across nearly two decades (trade mediator measurements from 2000–2018, for a total of 19 years) and recover estimates of the natural indirect and direct effects for each year. Figure 3 displays these estimates with error

bars indicating 95% confidence intervals for our estimates. Appendix Table 5 presents regression tables of the main model for each five-year increment in the two-decade period.² Across years, we observe evidence of both direct and indirect effects.

Consistent with H1, the natural indirect effects reflecting the pathway from regime type to voting via trade network interdependence are generally positive. The NIEs are statistically significantly different from zero in 2006, 2007, 2012, 2014–2018 (ranging from NIE=0.049 CI=[0.005, 0.093] in 2006 to NIE=0.104 CI=[0.027, 0.181] in 2018), indicating that increases in a country's level of democracy are associated with higher probabilities of voting “yes” to human rights resolutions primarily through shifts in its trade relations with other countries. These effect sizes are not only positive and significant, but also large, reflecting shifts in voting behavior between 0.136 and 0.295 standard deviations.

Consistent with H2, we also detect a natural direct effect of regime type on voting that is typically negative. The average point estimate across years is -0.09 , and the NDE is statistically significant and negative in 2000, 2005, 2006, 2007 and 2012 (ranging from NDE= -0.251 CI=[$-0.387, -0.116$] in 2000 to NDE= -0.148 CI=[$-0.273, -0.023$] in 2012), equivalent to shifts in voting behavior between -0.743 and -0.417 standard deviations. This suggests that, if democracy were increased while holding the trade network fixed, a state would lower their probability of voting “yes” on human rights resolutions, an effect plausibly attributable to strategic reticence or sovereignty concerns among established democracies. Read in light of the contested agenda described in Section 2, the negative direct effect is not evidence that democratization erodes concern for human rights. It indicates that, holding trade position fixed, more democratic states are less willing to lend their votes to the sovereignty-forward and developmental strand of the UNGA human rights agenda, consistent with accounts emphasizing sovereignty costs and limited marginal benefit from endorsing additional international commitments (Hafner-Burton *et al.*, 2015; Moravcsik, 2000). The positive indirect effect, by contrast, indicates that trade network position pulls states toward the positions favored by their partners on that same contested agenda.

Since the ATE is the sum of the (mostly positive) NIE and (mostly negative) NDE, the average effect of democracy can be seen as often “canceling out” and not appearing statistically significantly different from zero. The impacts of democracy channeled through trade networks can offset its direct impact on human rights voting. We interpret this pattern as reflecting competing mechanisms. The negative NDE is consistent with sovereignty and institutional constraints, creating more reason to hesitate in endorsing resolutions that might imply external scrutiny or binding oversight. Conversely, the positive NIE accords with an economic incentive and norm diffusion logic: states that expand trade ties or join preferential trade agreements and international trade organizations encounter material incentives and normative peer pressures through market access conditions, human-rights and labor clauses, and exposure to external actors, all of which raise the likelihood of voting yes on UNGA human rights resolutions. We find that semi- or weak democracies, which might lack deeply institutionalized human rights protections, still face these incentives and peer pressures when positioning themselves on UNGA human rights resolutions.

These results raise a further question: how do countries' positions in the latent trade network space relate to the mediation pathway, and what kinds of shifts in trade position could transmit the effects of democracy? We investigate the trade network's latent structure further in Figure 4, where we present visualizations of countries' trade positions as estimated from the trade network data (*i.e.*, their embeddings $\hat{X}$ as used in the network mediation model) across five-year increments in the two decades. We present estimated “trade space” locations of each country by taking the first and second principal components of the $\hat{X}$ matrix and aligning axes across the years. Countries are further colored by polyarchy score.

²For purposes of illustration and space considerations, we do not present every estimated model in table form, but all estimated models can be found in the open source replication files.

The x-axis (PC1) appears to capture variation consistent with overall economic size and influence within the trade network. High PC1 scores correspond to global trade hubs and large economies (e.g., the U.S., China, Germany and Japan), reflecting greater centrality and aggregate trade volume. The y-axis (PC2) captures the composition and diversification of states' trade portfolios. States with high PC2 values, such as New Zealand, Thailand and post-2010s Singapore, experience diversified trade across goods and partners, while states with low PC2 scores, such as Syria and Qatar, indicate concentration in a small set of commodities (most notably in hydrocarbons). PC2 also seems to reflect some orientation towards imports versus exports, with high scores associated with more balanced or import-oriented profiles and lower scores with strongly export-oriented profiles.

Illustrative country trajectories clarify these dimensions. China's PC1 score increased steadily over the two decades under study. In 2000, China still sat below the largest economic powerhouses such as U.S. and Japan (ranking 6th in PC1 score), but by 2018 it ranked #1 in PC1 score at 27.67. This shift reflects its transition toward manufactured goods and deeper integration into international supply chains through increased imports of intermediates and capital goods. The U.S.-China trade war beginning in 2018 led to a sharp decrease in China's direct exports to the U.S. and a reorientation towards alternative trading partners (UNCTAD, 2022).

The experience of Singapore, on the other hand, helps illustrate high and rising PC2. Its PC2 score rose from 2.46 (2000) to 5.76 (2018), with notable increases after 2010. As an import-dependent re-export hub, Singapore's diversification and orientation reflect deep participation in regional supply chains and an extensive and dense network of bilateral and regional FTAs—by 2014 roughly 74% of exports and 80% of imports were under preferential terms—contributing to its elevated PC2 score (World Trade Organization, 2016).

Contrastive cases further illuminate this trade network space. Syria exhibits a pronounced decline in PC1 (from 1.79 to -8.85) alongside an increase in PC2 (from -4.37 to -1.44) following the onset of conflict and sanctions after 2011: overall trade collapsed, with exports falling even more sharply than imports, shifting the economy to becoming more import-reliant (on refined fuel, medicine and food, for instance) during wartimes amid reductions in export capacity ("The Toll of War", n.d.). In contrast, the United Arab Emirates rose in PC1 values (from 3.72 in 2000 to 17.61 in 2018) and PC2 values (-5.31 to 2.87), consistent with active diversification strategies and greater centrality in global trade ("The present and future of global trade", 2020). Robustness checks confirm the substantive interpretation of PC2. Both export and import volumes are correlated with PC2 (see Appendix Figure 11) and regression analyses (see Appendix Table 8) indicate that import orientation—the share of imports in total trade—is associated with PC2, supporting the interpretation that PC2 captures diversification and import-export orientation.³

We can also use our estimated NIEs and NDEs together with the fitted network embeddings to simulate predicted (counterfactual) voting scenarios. To illustrate the natural indirect effect, we predict changes in a country's probability of voting "yes" when its trade network position is altered while the democracy score and other controls are held fixed. For these illustrative purposes, we focus on the year 2014. Consider Moldova and Japan. While both are democracies (Moldova's V-Dem score is 0.62, while Japan's is 0.82), Japan is a far larger and more central trading actor in comparison, with an export economy that is about 240 times larger than Moldova's, and with major trading partners drawn from countries like China, Thailand and the United States (compared to Moldova's main counterparts of Germany, Romania and Russia). While Japan has a competitive, high value-added and technology-intensive manufacturing sector, such as automobiles and precision machinery, Moldova's industrial base remains focused on agricultural and food processing, exporting relatively lower-margin goods which makes its economy less integrated into advanced global value

³These correlations are not mutually exclusive: PC2 is a latent coordinate summarizing position in the same underlying trade system, and it can correlate with export and import *levels* while also correlating with import *share*. The import-share association is a face-validity diagnostic for composition; it does not imply that PC2 is reducible to "imports only."

chains (“Moldova – Market Overview”, 2024). If Moldova were to “become more like” Japan in trade network position—that is, remaining at the same level of democracy but expanding the size and diversity of its trade—its predicted probability of voting yes on a human rights resolution would increase from 0.694 to 0.816.

This logic also carries through among countries that score lower in V-Dem polyarchy. Syria and Qatar score similarly low in V-Dem polyarchy, but Qatar maintains considerably larger and more diversified trade ties compared to Syria. While Syria’s main trade partners are drawn from countries like Jordan, Turkey and Saudi Arabia, Qatar’s major trading partners include China, Japan, Germany and the U.S. Our model indicates that if Syria were to take up a trade network position more akin to that of Qatar, its predicted likelihood of voting yes on a UNGA human rights resolution would increase from 0.83 to 0.975. Again, this may be unsurprising given related literature stating that when weak democracies seek to expand their trade market to more democratic countries, human rights conditionalities are frequently attached (Simmons & Elkins, 2004).

To further illustrate the natural direct effect, we simulate counterfactuals by varying a state’s democracy score while holding its trade network position and other controls constant. Across these counterfactual scenarios, increasing a state’s democracy score while holding trade position fixed reduces its predicted propensity to vote yes on UNGA human rights resolutions. For instance, if Saudi Arabia, which maintained lower average V-Dem scores throughout the 19 years (ranging from 0.015 to 0.019), were to “democratize” and take up a democracy score closer to that of Canada (0.812), its predicted likelihood of voting yes on a human rights resolution falls from 0.917 to 0.865. Similarly, if Nicaragua (2014 V-Dem score 0.321) were to transition towards a democracy that looked closer to Cyprus (2014 V-Dem score 0.845), this too would lower its predicted probability from 0.95 to 0.915.

The divergence between generally positive NIEs and more negative NDEs is most pronounced in 2006, 2007 and 2012. We attribute this amplification to contemporaneous shifts in international political-economic structure, domestic politics, and norm shocks that occurred around these periods of time. In the mid-2000s, the global economy was still in a growth phase punctuated by acute energy market disruption and price spikes, which heightened the politics of economic interdependence and the diplomatic stakes of market access and trade partner choices. These same years also saw protracted Middle East conflict (including the ongoing Iraq war and 2006 Lebanon war), which kept humanitarian and human rights framing unusually salient in multilateral diplomacy. The year 2012, by contrast, marks a post-global financial crisis environment in which economic vulnerability and adjustment remained pronounced (e.g., the eurozone crisis) alongside a reordered Middle East after the tumultuous Arab Spring, especially the escalation of the Syrian conflict, which concentrated international attention on sovereignty, intervention and rights-related UN agenda politics. The year also saw significant leadership transitions in several major powers (notably, in Russia, China, France, Mexico, Japan and the U.S.). Such turnovers can reset foreign-economic strategies and reallocate attention in multilateral fora like the UN.

These developments likely increased the relative importance of both economic incentives and normative signaling for UNGA voting. Heightened economic tensions make trade network position a more salient channel for foreign policy behavior, amplifying NIEs, while rapid domestic political changes might strengthen strategic or sovereignty-oriented motives that operate independently of trade, amplifying negative NDEs. Moreover, international normative mobilizations, such as transnational campaigns over atrocity prevention and civilian protection (including, for instance, heightened attention on the Darfur conflict in the mid-2000s), could have increased countries’ incentives to take public positions on the UN human rights agenda. This could have further modified the balance between direct and indirect effects of democracy in these years.

Robustness checks To probe the sensitivity of our results, we conduct several robustness checks. These are motivated by standard concerns about network orientation, outcome coding, agenda salience, and measurement validity in UNGA human rights resolutions. We summarize the resulting estimates in Appendix Table 5, reporting coefficients at five-year intervals (2000, 2005, 2010, 2015 and 2018; models that use the U.S. State Department “important resolution” control extend through 2017 only due to missing coverage for 2018 onward). Across these perturbations, the qualitative mediation pattern remains consistent: the direct effect of democracy is negative, while the indirect effect through trade networks is positive.

Appendix Table 5 reports alternative measurement and specification variants. The first column reports our main specification. Column (2) re-estimates the same model but constructs the mediator from the import trade network rather than the export network. This check addresses whether our conclusions depend on treating export-side economic embeddedness as the relevant channel, given that import dependence can generate distinct market-access and coalition pressures.⁴ Column (3) recodes the outcome as a binary indicator, probing robustness to a simpler measurement rule for abstention and non-response. Column (4) adds a control for whether a resolution was classified as “important” in the U.S. State Department’s Voting Practices in the United Nations, included as a pre-treatment resolution attribute where applicable, to reduce concerns that agenda salience rather than our mechanism drives patterns in voting. Column (5) is motivated most directly by potential concerns about the face validity of interpreting “yes” votes as “human rights support” in highly politicized UNGA items. As emphasized in the international relations literature, many resolutions related to human rights combine rights language with sovereignty-forward frames (e.g., “right to development” and “cultural diversity” agendas) or with country-specific diplomatic conflicts—most visibly Israel/Palestine—that can make a “yes” vote lean more towards coalitional or foreign policy signal rather than endorsement of human-rights protections per se. To address this concern, we implement a subsetting outcome construction that drops resolutions that we classify as belonging to two groups: (a) items centered on right to development and/or cultural diversity, and (b) items centered on Israel-Palestine.⁵ Although the fully subsetting specification is noisier, reflecting a thinner agenda and less stable estimation when many votes are removed, the sign pattern and substantive conclusions align with the main model: negative NDEs and positive NIEs. We push this exercise further in Appendix Tables 6 and 7, which drop each of six recurring and politically contested resolution categories in turn and then all six simultaneously, and recover the same pattern of negative direct and positive indirect effects, with the significant positive NIEs in 2015 and 2018 persisting throughout.

Finally, while we select the embedding dimension k using the cross-validation approach detailed in Section 3.2, we also inspect a larger range of possible choices of embedding dimension to determine whether our findings are sensitive to this choice. We present these in Appendix Figure 10. Our findings remain generally consistent across a range of embedding dimensions: the direct effects tend to be negative, whereas the indirect effects via the trade network tend to be positive for values of k chosen close to those specified in Section 3.2.

⁴Export- and import-defined adjacency matrices provide alternative orientations of the same bilateral trade flows, so the corresponding embeddings need not rank-order natural indirect effect magnitudes in every year. Year-to-year differences in the size of the NIE across these specifications are therefore best read as sensitivity to network orientation under shocks that can reweight import- versus export-side ties, rather than as a strict test of the export-stakes mechanism emphasized in the theory section. Across the robustness-check models in Appendix Table 5, the qualitative mediation pattern remains stable: negative NDEs and positive NIEs.

⁵All three authors independently reviewed the resolution universe and verified the removal process of the intended categories. We document the procedure and inclusion rules in Appendix Appendix 2.

5. Conclusion

This paper identifies international trade networks as a mechanism linking regime type to states' UNGA human-rights voting. We theorize that democratization alters the economic incentives for taking positions on UNGA human rights resolutions by changing trade interdependencies, producing both indirect effects that operate through network position and direct effects that operate independently of the network structure. Applying a recently-developed network mediation model by Hayes *et al.* (2025), we have demonstrated that democracy generates positive natural indirect effects via trade networks, but tends to produce negative natural direct effects. The opposing signs help reconcile prior empirical findings that report positive, null, or negative average effects of democracy on human-rights voting. These patterns are most pronounced in 2006, 2007 and 2012, years in which the international political economy was unusually turbulent, plausibly increasing the relative importance of both economic incentives routed through trade networks and the norm- and sovereignty-laden motives that shape UNGA voting.

Our contribution is threefold. First, we advance scholarship on the foreign-policy consequences of democratization by specifying a concrete, measurable pathway through which regime type shapes multilateral voting. Second, we extend research on trade and international organizations by demonstrating how network position conditions states signal in the UNGA. Third, we illustrate a causal identification strategy for networked mediators that embeds latent network structure within a potential-outcomes regression framework, offering applied researchers a principled approach for estimating direct and indirect effects when relational structure plays a mediating role.

Our findings carry implications for both the democratic backsliding literature and contemporary debates over global trade politics. In particular, the recent combination of U.S. trade wars and heightened tariff regimes provides a context in which our theoretical claims take on added salience. Our theory and findings imply that democratic backsliding, holding other factors constant, increases the likelihood that governments will vote yes on human rights resolutions in international fora such as the United Nations. States facing declining democracy scores may have incentives to take public positions on that agenda both to maintain international legitimacy and to offset reputational costs at home. Unlike entrenched democracies, which may perceive diminishing marginal benefits from endorsing such UNGA texts, backsliding regimes may retain reasons to vote yes on such resolutions. At the same time, this pattern of backsliding unfolds alongside heightened global trade uncertainty, shaped by U.S. protectionist policies, realignments among major economies, and intensified competition over raw materials (*Global Trade Update Facts and Figures: Trade Policy Uncertainty Looms over Global Markets*, 2025). These shifting economic ties interact with political incentives, underscoring the importance of theorizing trade networks as mediating channels in states' foreign policy behavior.

Finally, our study points to methodological opportunities for future scholarly work. We encourage future research to move toward theoretical frameworks and empirical strategies that incorporate network mechanisms where they naturally arise. Within international relations, political and economic networks are central to understanding how policies, economic decisions, and technological changes indirectly shape downstream political phenomena. Beyond this domain, the rise of social media as a dominant communication and information channel similarly suggests the potential for network-mediated pathways, whether in shaping party messaging, reinforcing organizational legitimacy, or consolidating political norms. Recognizing and modeling such mechanisms offers a path forward for more comprehensive theories and empirical findings. This suggests the need for network models that permit the application of causal inference tools in the presence of temporal dynamics, but this remains outside the scope of the present work.

6. Figures & Tables

Table 1. Example resolutions in the Human Rights (hr) issue category, categorized in the UNGA votes dataset (unvotes package). Resolutions in the table are randomly sampled from each year of the UNGA votes dataset, with two further resolutions sampled among *cultural diversity* and *right to development* resolutions to illustrate examples of each.

RCID	Year	Short	Description
4361	2001	HUMAN RIGHTS, IRAN	Situation of human rights in the Islamic Republic of Iran
4429	2002	HUMAN RIGHTS, GLOBALIZATION	Globalization and its impact on the full enjoyment of all human rights
4488	2003	Israeli practices affecting the human rights of the Palestin	Israeli practices affecting the human rights of the Palestinian people in the Occupied Palestinian Territory, including East Jerusalem : resolution / adopted by the General Assembly
4586	2004	Division for Palestinian Rights of the Secretariat : resolut	Division for Palestinian Rights of the Secretariat : resolution / adopted by the General Assembly
4616	2005	Global efforts for the total elimination of racism, racial d	Global efforts for the total elimination of racism, racial discrimination, xenophobia and related intolerance and the comprehensive implementation of and follow-up to the Durban Declaration and Programme of Action : resolution / adopted by the
4740	2006	RACIAL DISCRIMINATION, ELIMINATION	Inadmissibility of certain practices that contribute to fuelling contemporary forms of racism, racial discrimination, xenophobia and related intolerance : resolution / adopted by the General Assembly
4781	2007	PALESTINE QUESTION	Division for Palestine Rights of the Secretariat: resolution / adopted by the General Assembly
4882	2008	HUMAN RIGHTS, REPORTS	Situation of human rights in the Democratic People's Republic of Korea: resolution / adopted by the General Assembly
5030	2009	HUMAN RIGHTS, REPORTS, IRAN (ISLAMIC REPUBLIC OF)	Situation of human rights in the Islamic Republic of Iran : resolution / adopted by the General Assembly
5105	2010	HUMAN RIGHTS ADVANCEMENT	Promotion of peace as a vital requirement for the full enjoyment of all human rights by all : resolution / adopted by the General Assembly
5126	2011	HUMAN RIGHTS ADVANCEMENT	Human rights and cultural diversity : resolution / adopted by the General Assembly
5171	2011	TERRITORIES OCCUPIED BY ISRAEL–HUMAN RIGHTS	Applicability of the Geneva Convention relative to the Protection of Civilian Persons in Time of War, of 12 August 1949, to the Occupied Palestinian Territory, including East Jerusalem, and the other occupied Arab territories : resolution / adop
5239	2012	HUMAN RIGHTS ADVANCEMENT	Moratorium on the use of the death penalty : resolution / adopted by the General Assembly
5283	2013	TERRITORIES OCCUPIED BY ISRAEL–HUMAN RIGHTS	Applicability of the Geneva Convention relative to the Protection of Civilian Persons in Time of War, of 12 August 1949, to the Occupied Palestinian Territory, including East Jerusalem, and the other occupied Arab territories : resolution / adopted by the General Assembly
5476	2016	Combating glorification of Nazism, neo-Nazism and other practices that contribute to fuelling contemporary forms of racism, racial discrimination, xenophobia and related intolerance : resolution / adopted by the General Assembly	A/71/251 66a - Elimination of racism, racial discrimination, xenophobia and related intolerance. - RACIAL DISCRIMINATION–ELIMINATION
5483	2016	The right to development : resolution / adopted by the General Assembly	A/71/251 68b - Human rights questions, including alternative approaches for improving the effective enjoyment of human rights and fundamental freedoms. - HUMAN RIGHTS ADVANCEMENT;A/71/251 68b[11] - RIGHT TO DEVELOPMENT

Continued on next page

Table 1 – continued from previous page

RCID	Year	Short	Description
5623	2017	The occupied Syrian Golan : resolution / adopted by the General Assembly	A/72/251 54 - Report of the Special Committee to Investigate Israeli Practices Affecting the Human Rights of the Palestinian People and Other Arabs of the Occupied Territories. - TERRITORIES OCCUPIED BY ISRAEL–HUMAN RIGHTS
6113	2018	Israeli settlements in the Occupied Palestinian Territory, including East Jerusalem, and the occupied Syrian Golan : resolution / adopted by the General Assembly	A/73/251 55 - Report of the Special Committee to Investigate Israeli Practices Affecting the Human Rights of the Palestinian People and Other Arabs of the Occupied Territories. - TERRITORIES OCCUPIED BY ISRAEL–HUMAN RIGHTS
9084	2019	Work of the Special Committee to Investigate Israeli Practices Affecting the Human Rights of the Palestinian People and Other Arabs of the Occupied Territories	Work of the Special Committee to Investigate Israeli Practices Affecting the Human Rights of the Palestinian People and Other Arabs of the Occupied Territories

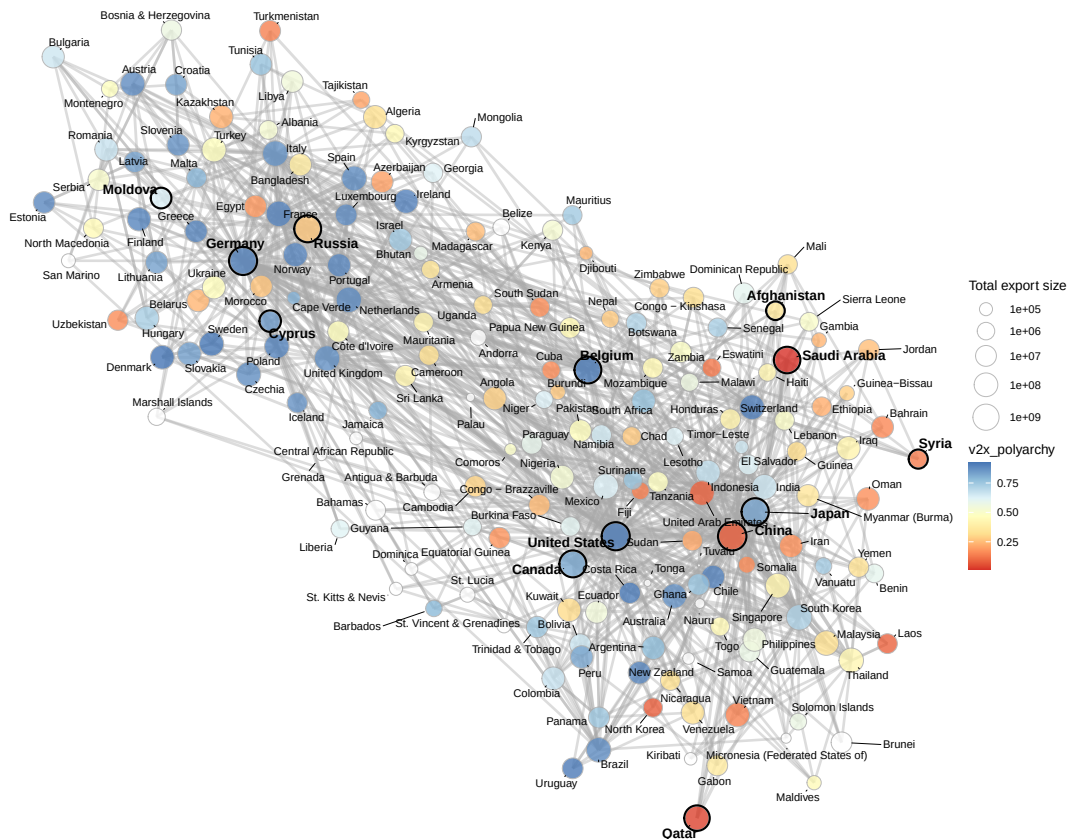


Figure 1. Illustration of trade network in 2014, drawn from the World Trade Flow Characterization (WTFC) dataset. Each node represents a country, and its size reflects the country's total export volume. Edges represent trade flows between countries. Node color indicates the country's level of democracy, measured by the V-Dem polyarchy score. For clarity, only the top five largest trade connections for each country are displayed. This plot demonstrates that democracy level alone does not perfectly predict the size of a country's trade volume, the strength of its trade connections, or its embeddedness in the global network. For instance, countries such as Qatar and Saudi Arabia, which have low democracy scores, appear to be as deeply embedded in the network as highly democratic states like the United States and Belgium. As states democratize, they can become more embedded in different trading communities (across the polyarchy spectrum).

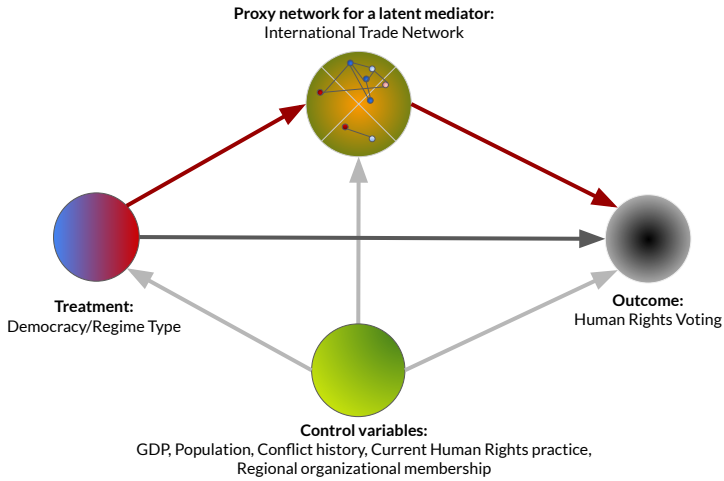


Figure 2. A directed acyclic graph (DAG) representing the causal pathways of latent mediation in a network. Control variables (C) in this model include all confounders. We are interested in the causal effect of a treatment (T) on an outcome (Y), both directly (dark gray arrow), as well as mediated by a proxy network (A) that captures the trade network latent position of each country (X) (red arrow pathway).

Acknowledgments We acknowledge the use of artificial intelligence tools in the writing of code associated with this article. All errors are entirely our own.

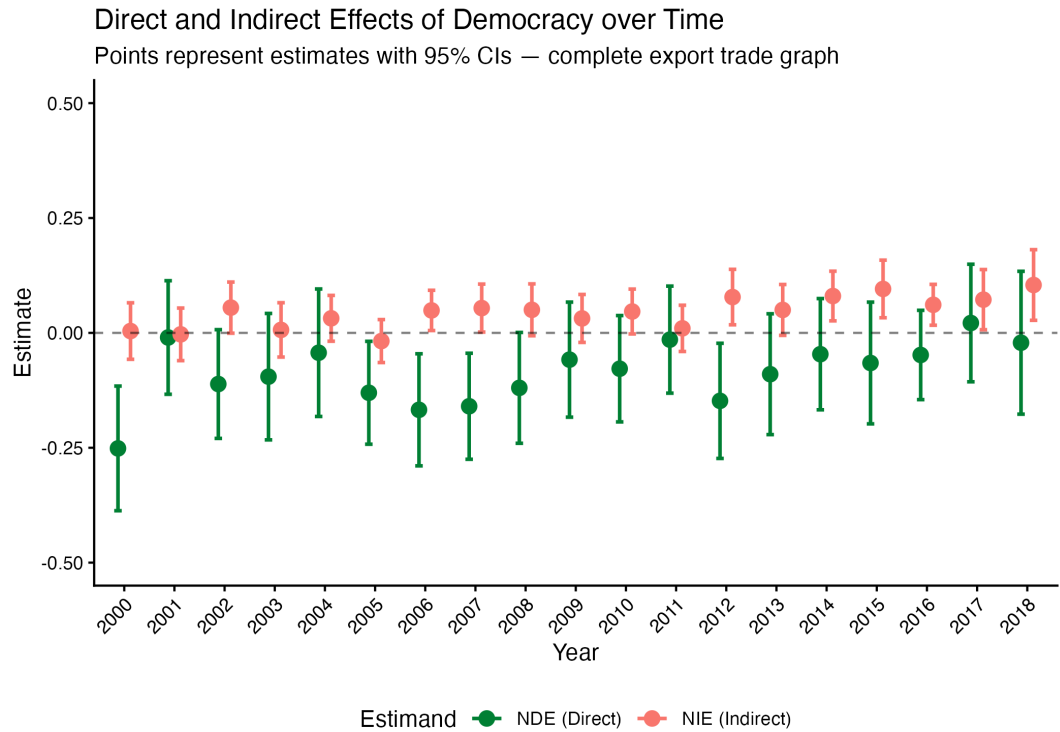


Figure 3. Natural Direct Effect (NDE) and Natural Indirect Effect (NIE) of Democracy on Human Rights Voting, using a time-lagged, fully aligned nodal dataset with a complete trade network graph. The x-axis plots the trade years of our study from 2000 to 2018, representing the periods when trade networks were measured. We employ a lagged model to ensure the temporal precedence of variables: treatment precedes the mediator, and the mediator precedes the outcome. Specifically, democracy is measured at $t - 1$, trade networks are measured at t , and voting outcomes are measured at $t + 1$. All other covariates such as logged GDP and population size are pre-treatment measurements taken at $t - 2$. The y-axis presents the estimated values for the Natural Indirect Effect (NIE) and Natural Direct Effect (NDE). Each coefficient point represents either the NIE (in orange) or the NDE (in green), with 95% confidence intervals. While not always statistically significant, the NDE generally yields negative coefficients, whereas the NIE shows positive coefficients. This pattern becomes more salient during the late 2000s compared to the early 2000s.

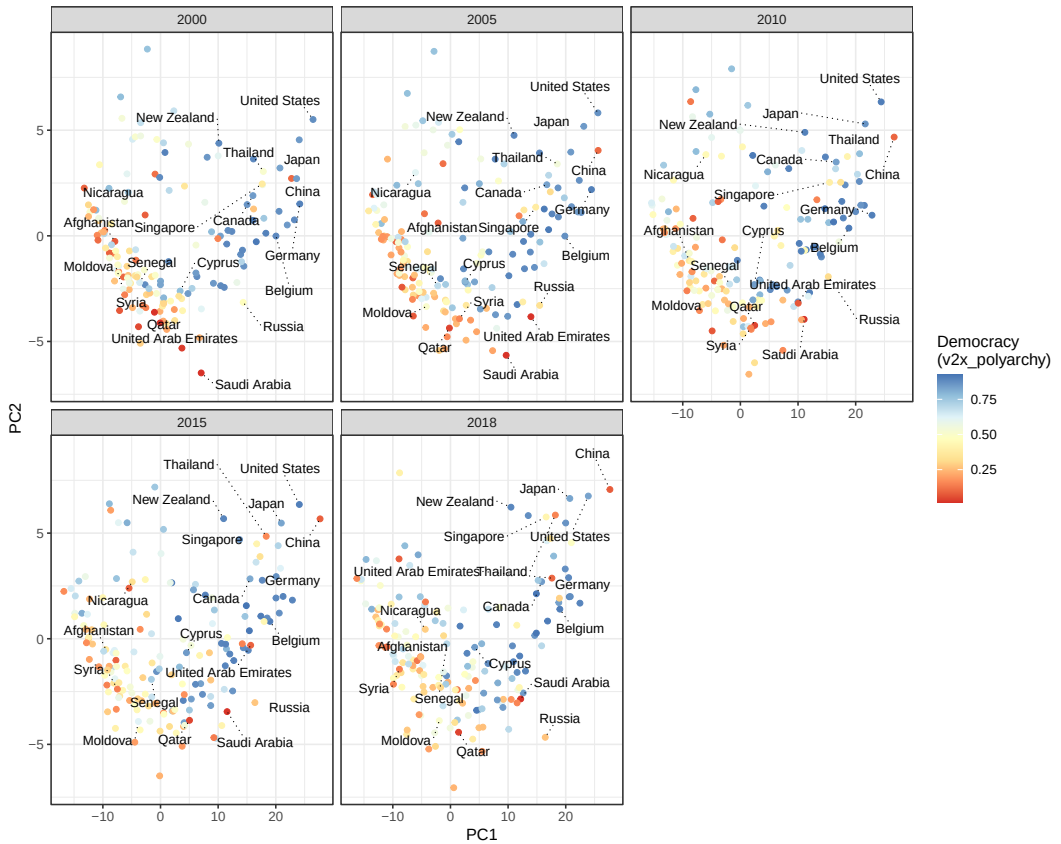


Figure 4. Principal Components of global trade-network adjacency matrices at five-year intervals (2000–2018). Second principal component (PC2) scores are sign-aligned across years using Grenada as a fixed reference. Because PCA axes are identified only up to sign, we align each year’s PC2 so Grenada’s score matches the reference-year orientation. Grenada is used because its PC2 loading is among the largest and most stable across the 2000–2018 panel.

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Data Availability Statement Replication data and code will be posted on Harvard Dataverse: `\url{https://doi.org/link}`.

Ethical Standards The research meets all ethical guidelines, including adherence to the legal requirements of the study country.

Author Contributions Y.R.: Conceptualization, Methodology, Investigation, Data Curation, Writing, Data Visualization. A.L.: Conceptualization, Methodology, Investigation, Writing, Supervision. K.L.: Methodology, Investigation, Writing, Supervision. All authors approved the final submitted draft.

Notes

References

- Appel, B. J., & Prorok, A. K. (2019). Third-Party Actors and the Intentional Targeting of Civilians in War. *British Journal of Political Science*, 49(4), 1453–1474. <https://doi.org/10.1017/S0007123417000175>
- Arel-Bundock, V. (2010, June 14). *WDI: World Development Indicators and Other World Bank Data*. Comprehensive R Archive Network. <https://doi.org/10.32614/CRAN.package.WDI>
- Athreya, A., Fishkind, D. E., Tang, M., Priebe, C. E., Park, Y., Vogelstein, J. T., Levin, K., Lyzinski, V., Qin, Y., & Sussman, D. L. (2018). Statistical Inference on Random Dot Product Graphs: A Survey. *Journal of Machine Learning Research*, 18(226), 1–92. Retrieved October 9, 2025, from <http://jmlr.org/papers/v18/17-448.html>
- Bailey, M. A., Strezhnev, A., & Voeten, E. (2017). Estimating dynamic state preferences from United Nations voting data. *Journal of Conflict Resolution*, 61(2), 430–456.
- Beer, C., & Mitchell, N. J. (2004). Democracy and Human Rights in the Mexican States: Elections or Social Capital? *International Studies Quarterly*, 48(2), 293–312. <https://doi.org/10.1111/j.0020-8833.2004.00302.x>
- Bernard, A., Jensen, J., & Schott, P. (2009). Importers, exporters and multinationals: A portrait of firms in the U.S. that trade goods. In *Producer Dynamics: New Evidence from Micro Data* (pp. 513–552). University of Chicago Press.
- Boehmer, C., Gartzke, E., & Nordstrom, T. (2004). Do intergovernmental organizations promote peace? *World politics*, 57(1), 1–38.
- Bogati, S., Firawati, T., Pinckney, J., & Thurber, C. (2025). How Civil Resistance Improves Inclusive Democracy. *Perspectives on Politics*, 23(2), 552–569. <https://doi.org/10.1017/S1537592724000999>
- Boockmann, B., & Dreher, A. (2011). Do human rights offenders oppose human rights resolutions in the United Nations? *Public Choice*, 146, 443–467.
- Boos, D. D., & Stefanski, L. A. (2013). *Essential statistical inference: Theory and methods* (Vol. 120). Springer Science & Business Media.
- Boucheron, S., Lugosi, G., & Massart, P. (2013). Concentration inequalities.
- Brazys, S., & Panke, D. (2017). Analysing Voting Inconsistency in the United Nations General Assembly. *Diplomacy & Statecraft*, 28(3), 538–560. <https://doi.org/10.1080/09592296.2017.1347450>
- Cao, X., Greenhill, B., & Prakash, A. (2013). Where is the tipping point? Bilateral trade and the diffusion of human rights. *British Journal of Political Science*, 43(1), 133–156.
- Carter, D. B., & Stone, R. W. (2015). Democracy and multilateralism: The case of vote buying in the UN General Assembly. *International Organization*, 69(1), 1–33.
- Chang, J. H., & Paul, S. (2024, June 10). *Embedding Network Autoregression for time series analysis and causal peer effect inference*. arXiv: 2406.05944 [stat]. <https://doi.org/10.48550/arXiv.2406.05944>
- Chen, F., Roch, S., Rohe, K., & Yu, S. (2021). *Estimating graph dimension with cross-validated eigenvalues*.
- Chen, F. R., Pevehouse, J. C. W., & Powers, R. M. (2023). Great Expectations: The Democratic Advantage in Trade Attitudes. *World Politics*, 75(2), 316–352. Retrieved October 11, 2025, from <https://muse.jhu.edu/pub/1/article/885485>
- Cinelli, C., Forney, A., & Pearl, J. (2024). A Crash Course in Good and Bad Controls. *Sociological Methods & Research*, 53(3), 1071–1104. <https://doi.org/10.1177/00491241221099552>
- Cingranelli, D., & Filippov, M. (2010). Electoral Rules and Incentives to Protect Human Rights. *The Journal of Politics*, 72(1), 243–257. <https://doi.org/10.1017/S0022381609990594>
- Coppedge, M., Gerring, J., Knutsen, C. H., Lindberg, S. I., Teorell, J., Altman, D., Angiolillo, F., Bernhard, M., Cornell, A., Gjerløw, H., Glynn, A., Grahm, S., Hicken, A., Kinzelbach, K., Marquardt, K., McMann, K., Mechkova, V., Medzihorsky, J., Neundorff, A., ... Sundström, A. (2025). *V-Dem Dataset V15*. Varieties of Democracy (V-Dem) Project. <https://doi.org/10.23696/VDEMDS25>
- Cottiero, C., & Haggard, S. (2023). Stabilizing Authoritarian Rule: The Role of International Organizations. *International Studies Quarterly*, 67(2), sqad031. <https://doi.org/10.1093/isq/sqad031>
- Davenport, C. (1999). Human rights and the democratic proposition. *Journal of Conflict Resolution*, 43(1), 92–116.
- Davenport, C., & Armstrong, D. A. (2004). Democracy and the violation of human rights: A statistical analysis from 1976 to 1996. *American Journal of Political Science*, 48(3), 538–554.
- Davis, C. L., & Pratt, T. (2021). The forces of attraction: How security interests shape membership in economic institutions. *The Review of International Organizations*, 16(4), 903–929. <https://doi.org/10.1007/s11558-020-09395-w>
- De Mesquita, B. B., Downs, G. W., Smith, A., & Cherif, F. M. (2005). Thinking inside the box: A closer look at democracy and human rights. *International Studies Quarterly*, 49(3), 439–457.
- Donnelly, J. (1999). Human rights, democracy, and development. *Human Rights Quarterly*, 21(3), 608–632.
- Dorussen, H., & Ward, H. (2008). Intergovernmental Organizations and the Kantian Peace: A Network Perspective. *Journal of Conflict Resolution*, 52(2), 189–212. <https://doi.org/10.1177/0022002707313688>
- Dorussen, H., & Ward, H. (2010). Trade networks and the Kantian peace. *Journal of Peace Research*, 47(1), 29–42. <https://doi.org/10.1177/0022343309350011>
- Dreher, A., & Sturm, J.-E. (2012). Do the IMF and the world bank influence voting in the UN general assembly? *Public Choice*, 151, 363–397.
- Emlinger, C., & Piton, S. (2014). *World trade flows characterization: Unit values, trade types and price ranges* (Working paper No. 2014–26). CEPII. http://www.cepii.fr/CEPII/en/bdd_modele/bdd_modele_item.asp?id=29
- Fariss, C. J. (2014). Respect for Human Rights has Improved Over Time: Modeling the Changing Standard of Accountability. *American Political Science Review*, 108(2), 297–318. <https://doi.org/10.1017/S0003055414000070>

- Gallop, M., & Minhas, S. (2021). A network approach to measuring state preferences. *Network Science*, 9(2), 135–152. <https://doi.org/10.1017/nws.2020.44>
- Gerring, J., Knutsen, C. H., & Berge, J. (2022). Does democracy matter? *Annual Review of Political Science*, 25, 357–375.
- Gleditsch, N. P., Wallensteen, P., Eriksson, M., Sollenberg, M., & Strand, H. (2002). Armed Conflict 1946–2001: A New Dataset. *Journal of Peace Research*, 39(5), 615–637. <https://doi.org/10.1177/0022343302039005007>
- Global Trade Update Facts and Figures: Trade Policy Uncertainty Looms over Global Markets. (2025). United Nations Trade and Development. https://unctad.org/system/files/official-document/ditcinf2025d7_en.pdf
- Gowa, J., & Mansfield, E. D. (1993). Power Politics and International Trade. *American Political Science Review*, 87(2), 408–420. <https://doi.org/10.2307/2939050>
- Greenhill, B. (2010). The company you keep: International socialization and the diffusion of human rights norms. *International studies quarterly*, 54(1), 127–145.
- Greenhill, B., Mosley, L., & Prakash, A. (2009). Trade-based Diffusion of Labor Rights: A Panel Study, 1986–2002. *American Political Science Review*, 103(4), 669–690. <https://doi.org/10.1017/S0003055409990116>
- Hafner-Burton, E. M. (2005). Trading human rights: How preferential trade agreements influence government repression. *International Organization*, 59(3), 593–629.
- Hafner-Burton, E. M. (2008). Sticks and stones: Naming and shaming the human rights enforcement problem. *International organization*, 62(4), 689–716.
- Hafner-Burton, E. M., Kahler, M., & Montgomery, A. H. (2009). Network Analysis for International Relations. *International Organization*, 63(3), 559–592. <https://doi.org/10.1017/S0020818309090195>
- Hafner-Burton, E. M., Mansfield, E. D., & Pevehouse, J. C. (2015). Human rights institutions, sovereignty costs and democratization. *British Journal of Political Science*, 45(1), 1–27.
- Hafner-Burton, E. M., & Montgomery, A. H. (2006). Power Positions: International Organizations, Social Networks, and Conflict. *Journal of Conflict Resolution*, 50(1), 3–27. <https://doi.org/10.1177/0022002705281669>
- Hagan, J. D. (1989). Domestic Political Regime Changes and Third World Voting Realignments in the United Nations, 1946–84. *International Organization*, 43(3), [xxxiii]–542. Retrieved March 24, 2025, from <https://heinonline.org/HOL/P?h=hein.journals/intorgz43&i=537>
- Hagan, J. D. (1994). Domestic Political Systems and War Proneness1. *Mershon International Studies Review*, 38, 183–207. <https://doi.org/10.2307/222714>
- Hathaway, O. A. (2002). The cost of commitment. *Stan. L. Rev.*, 55, 1821.
- Hathaway, O. A. (2007). Why Do Countries Commit to Human Rights Treaties? *Journal of Conflict Resolution*, 51(4), 588–621. <https://doi.org/10.1177/0022002707303046>
- Hayes, A., Fredrickson, M. M., & Levin, K. (2025, March 6). Estimating network-mediated causal effects via principal components network regression. arXiv: 2212.12041 [stat]. <https://doi.org/10.48550/arXiv.2212.12041>
- Hegre, H., Ellingsen, T., Gates, S., & Gleditsch, N. P. (2001). Toward a democratic civil peace? democracy, political change, and civil war, 1816–1992. *The American Political Science Review*, 95(1), 33–48. Retrieved October 11, 2025, from <http://www.jstor.org/stable/3117627>
- Hug, S., & Lukács, R. (2014). Preferences or blocs? Voting in the United Nations human rights council. *The review of international organizations*, 9, 83–106.
- Imai, K., Keele, L., & Tingley, D. (2010). A general approach to causal mediation analysis. *Psychological Methods*, 15, 309–334. <https://doi.org/10.1037/a0020761>
- Kim, I. S., & Milner, H. V. (2021). Multinational corporations and their influence through lobbying on foreign policy. The Brookings Institution. <https://www.brookings.edu/book/global-goliaths/>
- Kim, I. S., & Osgood, I. (2019). Firms in trade and trade politics. *Annual Review of Political Science*, 22(1), 399–417.
- Kim, S. Y., & Russett, B. (1996). The New Politics of Voting Alignments in the United Nations General Assembly. *International Organization*, 50(4), 629–652. Retrieved March 22, 2025, from <https://www.jstor.org/stable/2704240>
- Kim, Y., Kessler, D., & Levina, E. (2019). Graph-aware modeling of brain connectivity networks.
- Landman, T. (2005). *Protecting human rights: A comparative study*. Georgetown University Press.
- Lebovic, J. H., & Voeten, E. (2006). The politics of shame: The condemnation of country human rights practices in the unchr. *International Studies Quarterly*, 50(4), 861–888.
- Lebovic, J. H., & Voeten, E. (2009). The cost of shame: International organizations and foreign aid in the punishing of human rights violators. *Journal of Peace Research*, 46(1), 79–97.
- Levin, K., Lodhia, A., & Levina, E. (2022). Recovering shared structure from multiple networks with unknown edge distributions.
- Li, T., Levina, E., & Zhu, J. (2019). Prediction models for network-linked data. *The Annals of Applied Statistics*, 13(1), 132–164. <https://doi.org/10.1214/18-AOAS1205>
- Liu, H., Jin, I. H., Zhang, Z., & Yuan, Y. (2021). Social Network Mediation Analysis: A Latent Space Approach. *Psychometrika*, 86(1), 272–298. <https://doi.org/10.1007/s11336-020-09736-z>
- Lo, A., Olivella, S., & Imai, K. (2025). A Statistical Model of Bipartite Networks: Application to Cosponsorship in the United States Senate. *Political Analysis*, 1–20. <https://doi.org/10.1017/pan.2025.10021>
- Long, A. G., & Leeds, B. A. (2006). Trading for security: Military alliances and economic agreements. *Journal of Peace Research*, 43(4), 433–451.

- Lyzinski, V., Sussman, D. L., Tang, M., Athreya, A., & Priebe, C. E. (2014). Perfect clustering for stochastic blockmodel graphs via adjacency spectral embedding. *Electronic Journal of Statistics*, 8(2), 2905–2922. <https://doi.org/10.1214/14-EJS978>
- Maerz, S., Edgell, A., Hellemeier, S., Illchenko, N., & Fox, L. (2025, September 22). *Vdeminstitute/vdemdata*. Retrieved September 23, 2025, from <https://github.com/vdeminstitute/vdemdata>
- Mansfield, E. D., Milner, H. V., & Rosendorff, B. P. (2000). Free to trade: Democracies, autocracies, and international trade. *American political science review*, 94(2), 305–321.
- Mansfield, E. D., Milner, H. V., & Rosendorff, B. P. (2017). Why democracies cooperate more: Electoral control and international trade agreements. In *Global Trade* (pp. 215–252). Routledge.
- Mansfield, E. D., & Pevehouse, J. C. (2000). Trade blocs, trade flows, and international conflict. *International organization*, 54(4), 775–808.
- Mansfield, E. D., & Pevehouse, J. C. (2006). Democratization and international organizations. *International Organization*, 60(1), 137–167.
- Maoz, Z. (2009). The effects of strategic and economic interdependence on international conflict across levels of analysis. *American Journal of Political Science*, 53(1), 223–240.
- Maoz, Z., & Russett, B. (1993). Normative and structural causes of democratic peace, 1946–1986. *American political science review*, 87(3), 624–638.
- Milner, H. V., & Kubota, K. (2005). Why the Move to Free Trade? Democracy and Trade Policy in the Developing Countries. *International Organization*, 59(1), 107–143. <https://doi.org/10.1017/S002081830505006X>
- Moldova – Market Overview [publisher: International Trade Administration]. (2024, March). Retrieved October 11, 2025, from <https://www.trade.gov/country-commercial-guides/moldova-market-overview>
- Moravcsik, A. (2000). The Origins of Human Rights Regimes: Democratic Delegation in Postwar Europe. *International Organization*, 54(2), 217–252. <https://doi.org/10.1162/002081800551163>
- Mosler, M., & Potrafke, N. (2020). International political alignment during the Trump presidency: Voting at the UN general assembly. *International Interactions*, 46(3), 481–497. <https://doi.org/10.1080/03050629.2020.1719405>
- Neyman, J. (1923). On the application of probability theory to agricultural experiments: Essay on principles [Original Polish publication, English translation commonly cited from Statistical Science reprint (1990).]. *Annals of Agricultural Sciences*, 10(1), 1–51. <https://doi.org/10.1214/ss/1177011933>
- Onderco, M. (2014). “Tell Me What You Want”: Analyzing the Visegrád Countries’ Votes in the UN General Assembly. *East European Politics and Societies*, 28(1), 63–83. <https://doi.org/10.1177/0888325413484757>
- Pevehouse, J., & Russett, B. (2006). Democratic international governmental organizations promote peace. *International organization*, 60(4), 969–1000.
- Prasad, S. K., & Nooruddin, I. (2024). States living in glasshouses ... : Why fighting domestic insurgency changes how countries vote in the UN human rights council. *Conflict Management and Peace Science*, 41(5), 556–573. <https://doi.org/10.1177/07388942231198489>
- The present and future of global trade: Technology and ICT will fuel the future of trade and Dubai is well placed to be ahead of the curve. [publisher: Dubai Department of Economy and Tourism]. (2020, October). <https://www.investindubai.gov.ae/en/insights-and-resources/news-insights/present-future-global-trade>
- Primiano, C. B., & Xiang, J. (2016). Voting in the UN: A second image of China’s human rights. *Journal of Chinese Political Science*, 21, 301–319.
- Relión, J. D. A., Kessler, D., Levina, E., & Taylor, S. F. (2019). Network classification with applications to brain connectomics. *The annals of applied statistics*, 13(3), 1648.
- Robinson, D. (2016, July 8). *Unvotes: United Nations General Assembly Voting Data*. Comprehensive R Archive Network. <https://doi.org/10.32614/CRAN.package.unvotes>
- Rubin, D. B. (2005). Causal inference using potential outcomes: Design, modeling, decisions. *Journal of the American statistical Association*, 100(469), 322–331.
- Russett, B., Layne, C., Spiro, D. E., & Doyle, M. W. (1995). The Democratic Peace. *International Security*, 19(4), 164–184. Retrieved June 9, 2023, from <https://muse.jhu.edu/pub/6/article/447382>
- Russett, B. M., & Oneal, J. R. (2001). *Triangulating peace : Democracy, interdependence, and international organizations*. W.W. Norton. Retrieved March 24, 2025, from <https://cir.nii.ac.jp/crid/1130282270528459648>
- Simmons, B. A. (2009). Mobilizing for human rights: International law in domestic politics. In *Leading works in international law* (pp. 177–188). Routledge.
- Simmons, B. A., & Elkins, Z. (2004). The Globalization of Liberalization: Policy Diffusion in the International Political Economy. *American Political Science Review*, 98(1), 171–189. <https://doi.org/10.1017/S0003055404001078>
- Sussman, D. L., Tang, M., Fishkind, D. E., & Priebe, C. E. (2012). A Consistent Adjacency Spectral Embedding for Stochastic Blockmodel Graphs. *Journal of the American Statistical Association*, 107(499), 1119–1128. <https://doi.org/10.1080/01621459.2012.699795>
- Sussman, D. L., Tang, M., & Priebe, C. E. (2014). Consistent Latent Position Estimation and Vertex Classification for Random Dot Product Graphs. *IEEE Transactions on Pattern Analysis and Machine Intelligence*, 36(1), 48–57. <https://doi.org/10.1109/TPAMI.2013.135>
- Teorell, J., & Lindberg, S. I. (2019). Beyond Democracy–Dictatorship Measures: A New Framework Capturing Executive Bases of Power, 1789–2016. *Perspectives on Politics*, 17(1), 66–84. <https://doi.org/10.1017/S1537592718002098>

The Toll of War: The Economic and Social Consequences of the Conflict in Syria. (n.d.). Retrieved October 11, 2025, from <https://www.worldbank.org/en/country/syria/publication/the-toll-of-war-the-economic-and-social-consequences-of-the-conflict-in-syria>

UNCTAD. (2022). *China's structural transformation: What can developing countries learn?* United Nations Conference on Trade; Development.

Van der Vaart, A. W. (2000). *Asymptotic statistics* (Vol. 3). Cambridge university press.

VanderWeele, T. (2015, February 13). *Explanation in Causal Inference: Methods for Mediation and Interaction*. Oxford University Press.

Voeten, E. (2013). Data and analyses of voting in the united nations: General assembly. *Routledge handbook of international organization*, 54–66.

Vreeland, J. R. (2008). Political institutions and human rights: Why dictatorships enter into the united nations convention against torture. *International Organization*, 62(1), 65–101.

World Trade Organization. (2016). Trade policy review: Singapore (2016) [Accessed: October 1, 2026]. *World Trade Organization*.

Yasar, M. (2013). Political influence of exporting and import-competing firms: Evidence from eastern european and central asian countries. *World Development*, 51, 154–168.

Appendix 1. Descriptive summaries

Table 2. Summary statistics of nodal variables. For character variables (country and year), N indicates the number of unique values. The final dataset covers 19 years and 190 countries in total. For detailed statistics on each year's dataset and the number of countries included, see *N Countries* in Table 3.

Variable	Summary Statistics			
	Min	Mean	Max	N
Year IV (democracy measured, year t)	1999	-	2017	19
Country	-	-	-	190
V-Dem Polyarchy Score	0.01	0.53	0.92	3139
Log GDP (scaled)	-3.02	-0.03	2.92	3402
Log Population (scaled)	-3.03	0	2.59	3426
Human Rights Index (scaled)	-2.85	-0.01	1.13	3098
Interstate Conflict Count (year t-1)	0	0.38	8	3426
Intrastate Conflict Count (year t-1)	0	0.19	6	3426
Interstate Conflict Count (year t-2)	0	0.36	8	3411
Intrastate Conflict Count (year t-2)	0	0.19	6	3411

Composition of the human-rights agenda. A natural concern is that our estimates could be driven by a particular slice of the agenda rather than by voting yes on human rights resolutions broadly. If certain kinds of resolutions attract distinctive patterns of state attention, they could exert undue influence on the results. We therefore identify six categories that are both highly recurring and politically contested, in the sense that their rights language is bound up with a separate political dispute likely to structure how states position themselves. These are also the categories most often cited as evidence that an affirmative vote is not a clean signal of a state's rights commitments. The characterizations below draw on the resolution texts and short subject labels used for classification, on the observed roll-call record, and on prior accounts of these agenda items (Boockmann & Dreher, 2011; Primiano & Xiang, 2016). The two clearest cases are also the two smallest: Globalization and Nazism. Resolutions on globalization and its impact on the full enjoyment of human rights are routinely opposed by the states conventionally regarded as most rights-committed, while supported by China, Belarus, Cuba, the DPRK, Iran and Saudi Arabia; a “yes” here tracks a developmental and sovereignty-forward reading of rights rather than a liberal one. The annual resolution on combating glorification of Nazism is treated by the United States, EU member states and the Nordic

Table 3. Year structure used in the analysis. Compared to the year (*t*) when trade networks are measured, democracy is measured in a prior year (*t* − 1), covariates are measured one year before the democracy score (*t* − 2), and UNGA votes are measured one year after the trade network (*t* + 1).

Summary of Year Structure Used in Analysis				
Covariates (t-2)	Democracy (t-1)	Trade Network (t)	UNGA Votes (t+1)	N Countries
1998	1999	2000	2001	173
1999	2000	2001	2002	182
2000	2001	2002	2003	180
2001	2002	2003	2004	186
2002	2003	2004	2005	180
2003	2004	2005	2006	187
2004	2005	2006	2007	188
2005	2006	2007	2008	186
2006	2007	2008	2009	187
2007	2008	2009	2010	188
2008	2009	2010	2011	189
2009	2010	2011	2012	187
2010	2011	2012	2013	189
2011	2012	2013	2014	189
2012	2013	2014	2015	187
2013	2014	2015	2016	190
2014	2015	2016	2017	189
2015	2016	2017	2018	190
2016	2017	2018	2019	189

Distribution of V-Dem Polyarchy Score

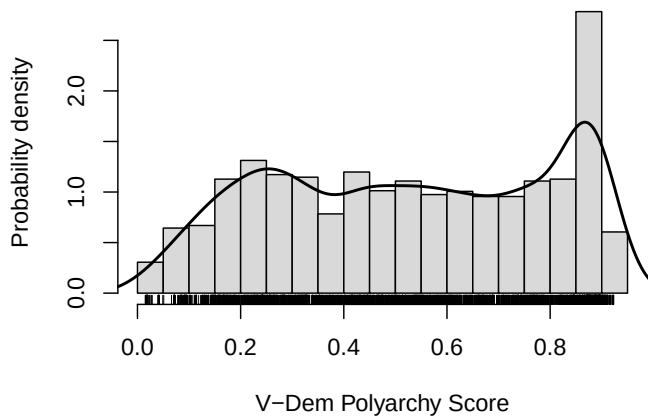


Figure 5. Distribution of independent variable, democracy index from V-dem project (v2x_polyarchy).

countries—which oppose or abstain, particularly after 2014—as Russian propaganda, and is supported by Russia, China and Venezuela. Resolutions on racial discrimination and the Durban follow-up, and

the large bloc of Israel/Palestine items, are similarly recurring and politicized; we group the occupied Syrian Golan with the latter because the Assembly takes these under a common agenda item and some texts cover both territories. The right to development and human rights and cultural diversity round out the set, both long-standing vehicles for a sovereignty-forward conception of the rights regime. What unites the six is not that an affirmative vote on them is uninformative, but that it is informative about something other than domestic rights practice, which is precisely the contested positioning our theory takes as its object. The residual *Others* category is large by construction rather than through classification failure. Whether these categories in fact drive our estimates is the question we take up in Appendix Tables 6–7.

Table 4. Composition of the 430 UNGA human-rights resolutions underlying the outcome, pooled across vote years 2001–2019. The counting unit is a distinct roll call (*rcid*), not a country-vote: each resolution is voted on by roughly 190 states but counted once. *N Years Present* is the number of the 19 vote years in which a category appears. Categories are assigned by case-insensitive keyword matching on each resolution’s subject label (*short*) and description (*descr*). Because they are not naturally disjoint, each resolution is assigned to the first category it matches in the order listed—Nazism precedes racial discrimination because its title also names racism, racial discrimination and xenophobia—and is counted exactly once. Resolutions matching no category are reported as *Others*.

Type	N Resolutions	N Years Present	% of Total
Globalization and human rights	18	17	4.2
Combating glorification of Nazism	8	8	1.9
Racial discrimination and Durban	28	18	6.5
Right to development	18	18	4.2
Human rights and cultural diversity	6	6	1.4
Israel/Palestine	124	19	28.8
Others	228	19	53.0
Total	430	19	100.0

Appendix 2. Main model and robustness checks

This Section presents a series of robustness checks on our main findings. We first report a regression table (Table 5) with the main model and four robustness specifications side by side (evaluated at five-year intervals—2000, 2005, 2010, 2015, 2018—for space considerations); each robustness column is paired below with a corresponding figure that plots the same specification's NDE and NIE estimates across the full time series (Figures 6, 7, 8, and 9). Note that y-axis scales are not uniform across these figures: Figure 6 uses the same $[-0.5, 0.5]$ range as the main results (Figure 3) for direct comparability, while the remaining robustness figures use a wider $[-1, 1]$ range. We then assess sensitivity to the choice of embedding dimension (Figure 10), plotting direct and indirect effects across ranks 2–80 at five-year increments against the rank selected by our cross-validation procedure. Finally, we extend this exercise to the full set of six recurring and politicized resolution categories from Table 4, re-estimating the main specification after dropping each individually and then all six together (Tables 6–7). For all regression estimations conducted, we utilize the main model specification as a baseline to start off, and then adjust based off of the robustness check (e.g. if we are testing robustness to different ways of measuring the outcome, we utilize the same specification and change the DV).

Across all of these specifications, the estimates track the main model closely: direct effects are negative and significant in 2000 and 2005 and attenuate toward zero thereafter, indirect effects are positive from 2010 onward, and the significant positive NIEs in 2015 and 2018 persist even when all six politicized categories are removed, though with wider intervals given the thinner agenda. We therefore find no indication that the mediation pattern we report is an artifact of any particular category of resolutions, nor of the specific choices made in measuring the mediator, the outcome, or agenda salience.

- **Table 5** (main + 4 robustness columns, selected years):
 - Column (1): main model specification.
 - Column (2): import trade network mediator instead of export. *Full time series: Figure 6.*
 - Column (3): binary outcome `vote_yes` instead of `vote_n`. *Full time series: Figure 7.*
 - Column (4): adds `importantvote` control. *Full time series: Figure 8.*
 - Column (5): drops Israel/Palestine, Right to Development, and Cultural Diversity resolutions. *Full time series: Figure 9.*
- **Figure 10**: sensitivity of NDE/NIE to embedding dimension, ranks 2–80, five-year increments.
- **Tables 6–7**: test sensitivity to each of the six recurring/politicized resolution categories from Table 4 individually (Cols 2–7: Globalization, Nazism, Racism/Durban, Right to Development, Cultural Diversity, Israel/Palestine), then all six together (Col 8)—a broader version of the exclusion underlying Column (5) of Table 5, which drops only three of the six (Right to Development, Cultural Diversity, Israel/Palestine).

Table 5. Robustness check: NDE/NIE estimates for democracy (v2x_polyarchy) at best K across model specifications ($k = 10, 12, 11, 12, 11$, respectively; chosen using the approach in Chen et al. (2021)), selected trade years (2000, 2005, 2010, 2015, 2018). The first model (1) is the main model (which is estimated with the export trade graph). Robustness check models (2)–(5) use an import trade graph, an alternative binary DV, an important vote control, and a restricted sample that excludes all resolutions concerning highly politicized issues (Israel/Palestine, right to development, and cultural diversity), respectively. All models include the set of covariate controls from the main model, measured in the year preceding the democracy variable (V-Dem score): scaled GDP and population, human rights index, WTO/EU/OIC membership status, and the presence of inter-state and intra-state conflicts (measured in the covariate year and the year prior). Resolution fixed effects are also included.

Year		(1) Main	(2) Import Network	(3) Binary DV	(4) Imp. Vote	(5) Subset FE
2000	NDE	-0.251*** [-0.387, -0.116]	-0.255*** [-0.383, -0.128]	-0.278*** [-0.463, -0.094]	-0.254*** [-0.389, -0.119]	-0.269*** [-0.472, -0.066]
	NIE	0.004 [-0.057, 0.065]	0.008 [-0.033, 0.048]	-0.001 [-0.085, 0.083]	0.005 [-0.056, 0.067]	-0.015 [-0.109, 0.078]
	NDE	-0.130*** [-0.242, -0.018]	-0.134*** [-0.241, -0.026]	-0.097 [-0.248, 0.054]	-0.130*** [-0.240, -0.020]	-0.193*** [-0.342, -0.044]
2005	NDE	-0.018 [-0.065, 0.029]	-0.014 [-0.051, 0.023]	-0.014 [-0.077, 0.048]	-0.019 [-0.065, 0.027]	-0.039 [-0.102, 0.025]
	NIE	-0.078 [-0.194, 0.038]	-0.055 [-0.167, 0.057]	0.006 [-0.143, 0.156]	-0.066 [-0.182, 0.049]	-0.058 [-0.222, 0.106]
	NDE	0.047 [-0.002, 0.095]	0.024 [-0.016, 0.064]	0.053 [-0.010, 0.115]	0.053*** [0.004, 0.101]	0.079*** [0.008, 0.149]
2010	NDE	-0.066 [-0.198, 0.067]	-0.038 [-0.166, 0.090]	-0.056 [-0.235, 0.123]	-0.073 [-0.209, 0.064]	-0.089 [-0.261, 0.084]
	NIE	0.096*** [0.033, 0.158]	0.068*** [0.012, 0.124]	0.140*** [0.055, 0.225]	0.097*** [0.032, 0.162]	0.129*** [0.047, 0.211]
	NDE	-0.021 [-0.177, 0.134]	-0.038 [-0.189, 0.112]	-0.003 [-0.211, 0.206]	NA	-0.019 [-0.196, 0.157]
2015	NDE	0.104*** [0.027, 0.181]	0.121*** [0.056, 0.187]	0.129*** [0.027, 0.230]	NA	0.133*** [0.045, 0.221]
	NIE					
	NDE					
2018	NDE					
	NIE					
	NDE					
2020	NDE					
	NIE					
	NDE					

Trade network	Export	Import	Export	Export	Export
Dependent variable	Numeric	Numeric	Binary	Numeric	Numeric
Important vote ctrl	No	No	No	Yes	No
Resolution subset	All	All	All	All	FE only
Controls	Yes	Yes	Yes	Yes	Yes

*** 95% CI excludes zero ($p < 0.05$, two-sided).

Issue-exclusion robustness. A potential source of concern is that human-rights voting patterns may differ systematically for particularly salient or substantively distinct issue types. To evaluate this possibility, we construct subsetting samples that remove (i) resolutions related to Israel/Palestine and (ii) resolutions targeting the Right to Development and Cultural Diversity. The removals are implemented using a pre-specified keyword filter, followed by manual screening: all three authors independently reviewed hundreds of resolutions to verify that each excluded item fell within the intended categories and that borderline cases were handled consistently. We further extend this test to the remaining three recurring and contested categories identified in Table 4—Globalization, Nazism, and Racial Discrimination/the Durban follow-up—evaluating the effects of excluding each of the six categories individually and all six jointly in Appendix Tables 6–7. These exclusion specifications are reported in Appendix Tables 5 (Column 5), 6 and 7.

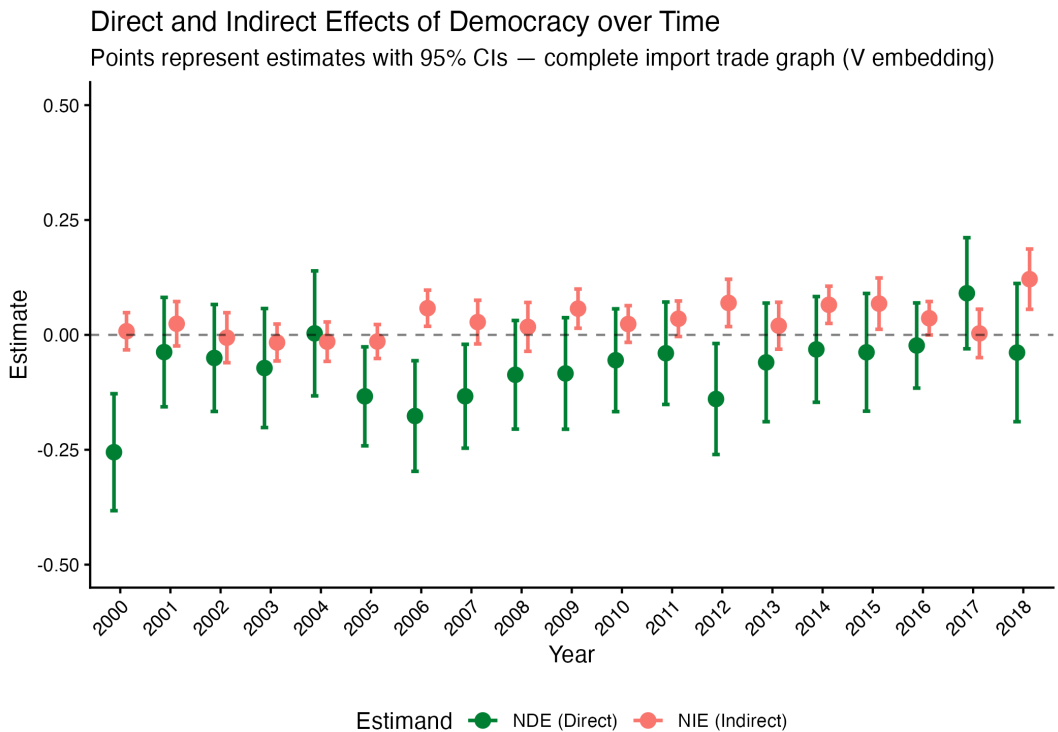


Figure 6. Natural Direct Effect (NDE) and Natural Indirect Effect (NIE) of Democracy on Human Rights Voting through the Import Trade Network: Full time-series estimates using the specification in Table 5, Column (2). Estimated NDE (green) and NIE (orange) of democracy on human rights voting behavior for each year in our data set, with error bars indicating 95% confidence intervals. Estimates are based on the same network mediation model as the main analysis, using *import* trade flows to construct the trade network mediator. The overall pattern is consistent with the main results (Figure 3): NDEs are generally negative, while NIEs are generally positive. However, both effects are attenuated toward zero and show noisier pathways, indicating that the estimated effects are weaker and less consistently statistically significant when the trade network is constructed from import rather than export trade flows.

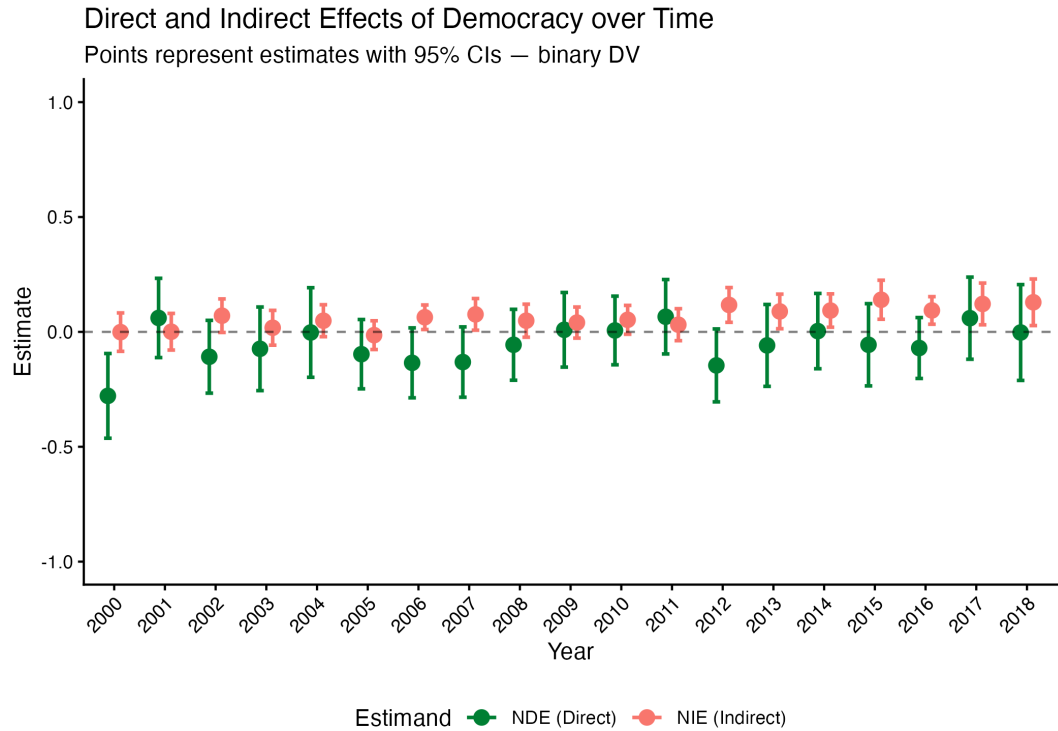


Figure 7. Natural Direct Effect (NDE) and Natural Indirect Effect (NIE) of Democracy on Human Rights Voting, Using an Alternative Voting Measurement: Full time-series estimates using the specification in Table 5, Column (3). Estimated NDE (green) and NIE (orange) of democracy on human rights voting behavior for each year in our data set, with error bars indicating 95% confidence intervals. Estimates are based on the same network mediation model as the main analysis, using an alternative outcome measure, *vote_yes*, a binary variable that codes abstention as 0 rather than 0.5. This alternative outcome measure does not materially change the overall pattern of results: NIEs are consistently positive from 2012 to 2018, while NDEs are generally negative across the years. The negative pattern of NDEs is consistent with the main results shown in Figure 3, although the estimates are not statistically significant in most years.

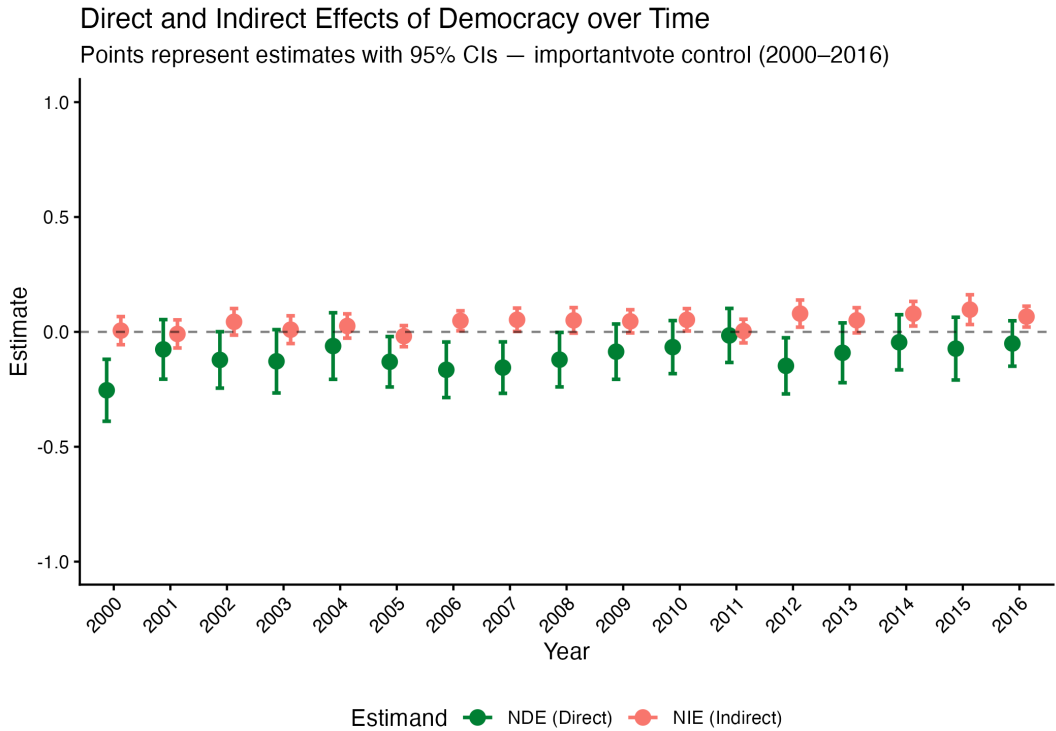


Figure 8. Natural Direct Effect (NDE) and Natural Indirect Effect (NIE) of Democracy on Human Rights Voting, Including Indicator for Important Votes: Full time-series estimates using the specification in Table 5, Column (4). Estimated NDE (green) and NIE (orange) of democracy on human rights voting behavior for each year in our data set, with error bars indicating 95% confidence intervals. Estimates are based on the same network mediation model as the main analysis, with an *importantvote* variable added to the main model. Because UNGA votes do not include the *importantvote* measure from 2018 onward, the plot is limited to 2000–2016, the years for which the measure is available in our data. The overall pattern is consistent with the main results (Figure 3): NDEs are typically negative, while NIEs are typically positive.

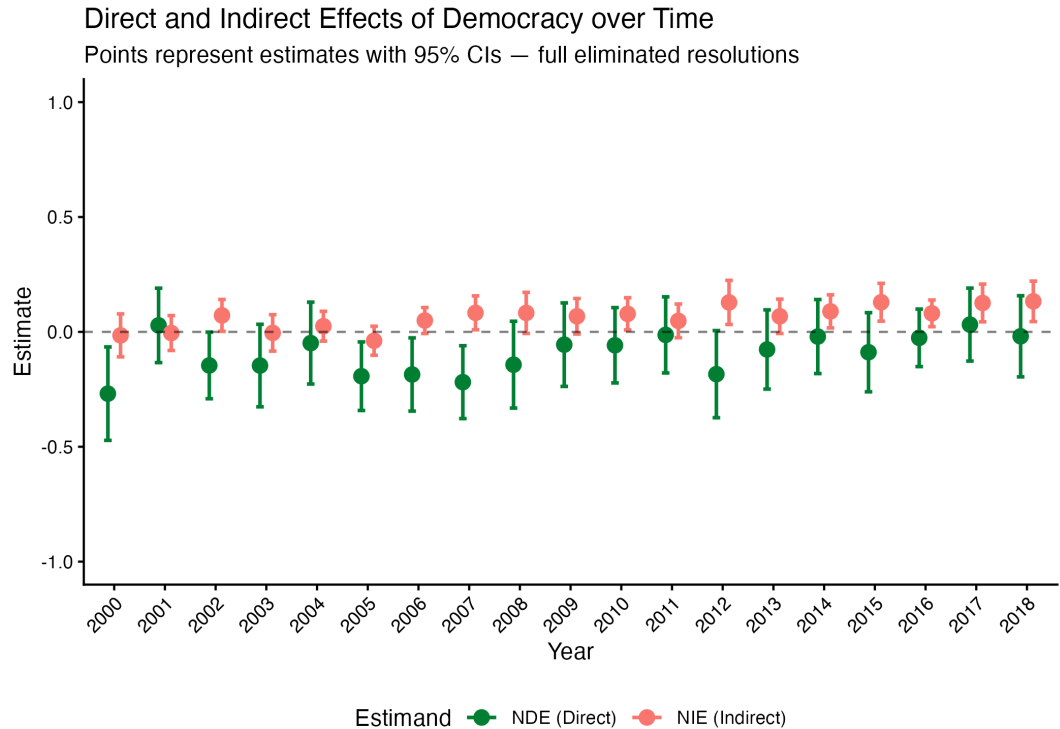


Figure 9. Natural Direct Effect (NDE) and Natural Indirect Effect (NIE) of Democracy on Human Rights Voting, Subsetting Out Politicized Resolutions: Full time-series estimates using the specification in Table 5, Column (5). Estimated NDE (green) and NIE (orange) of democracy on HR voting behavior, for each of the years in our data set, with error bars indicating 95% confidence intervals. Estimates are based on the same network mediation model as the main analysis, using a dependent variable constructed from the subset of human-rights roll calls that excludes resolutions on *Israel/Palestine* and resolutions centered on the *right to development* and *cultural-diversity* frames, following the exclusion rules described in the text and appendix. Subsetting does not materially change the overall pattern from the main result (Figure 3): NDEs are typically negative and NIEs typically positive relative to the main (unrestricted) results.

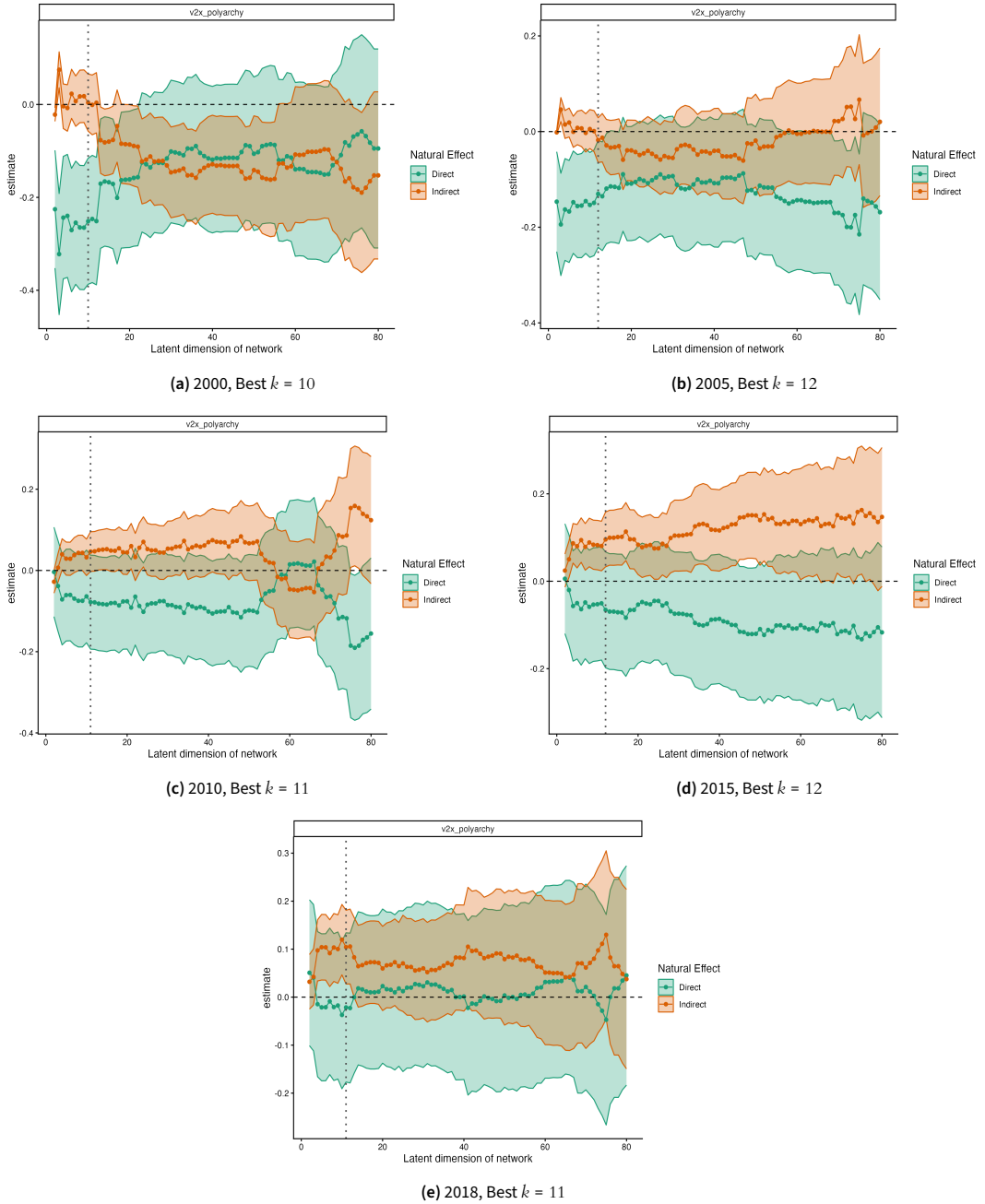


Figure 10. Sensitivity to rank. Direct and indirect effects (via trade networks) of democracy on human-rights-related UNGA votes across all latent dimensions (ranks 2–80) for five-year increments. Each year corresponds to the measurement of the trade network, while the UNGA vote (outcome) is recorded one year later (2001, 2006, 2011, 2016, and 2019). Gray vertical dotted lines indicate the optimal rank selected using the best-rank selection method.

Table 6. Sensitivity of the democracy (v2x_polyarchy) NDE/NIE estimates to each recurring and potentially politicized resolution category (see Table 4 for category definitions; continued in Table 7), selected years (2000, 2005, 2010, 2015). Column (1) is the main model on the full 430-resolution sample. Columns (2)–(4) re-estimate the same model after dropping one category’s resolutions at a time: globalization and human rights, combating glorification of Nazism, and racial discrimination and Durban follow-up, respectively. All columns include the same covariate controls and resolution fixed effects as the main model.

Year	(1) Main	(2) Excl. Globalization	(3) Excl. Nazism	(4) Excl. Racism/Durban
2000	NDE -0.251*** [-0.387, -0.116]	-0.225*** [-0.363, -0.087]	-0.251*** [-0.387, -0.116]	-0.251*** [-0.387, -0.116]
	NIE 0.004 [-0.057, 0.065]	0.009 [-0.053, 0.071]	0.004 [-0.057, 0.065]	0.004 [-0.057, 0.065]
2005	NDE -0.130*** [-0.242, -0.018]	-0.098 [-0.212, 0.016]	-0.130*** [-0.242, -0.018]	-0.130*** [-0.249, -0.011]
	NIE -0.018 [-0.065, 0.029]	-0.010 [-0.056, 0.037]	-0.018 [-0.065, 0.029]	-0.018 [-0.068, 0.031]
2010	NDE -0.078 [-0.194, 0.038]	-0.064 [-0.181, 0.054]	-0.078 [-0.194, 0.038]	-0.082 [-0.206, 0.042]
	NIE 0.047 [-0.002, 0.095]	0.038 [-0.011, 0.088]	0.047 [-0.002, 0.095]	0.045 [-0.007, 0.097]
2015	NDE -0.066 [-0.198, 0.067]	-0.048 [-0.184, 0.087]	-0.057 [-0.196, 0.082]	-0.057 [-0.196, 0.081]
	NIE 0.096*** [0.033, 0.158]	0.080*** [0.017, 0.144]	0.092*** [0.027, 0.158]	0.092*** [0.027, 0.157]
2018	NDE -0.021 [-0.177, 0.134]	-0.021 [-0.177, 0.134]	-0.003 [-0.169, 0.162]	-0.012 [-0.177, 0.154]
	NIE 0.104*** [0.027, 0.181]	0.104*** [0.027, 0.181]	0.106*** [0.025, 0.188]	0.105*** [0.023, 0.186]

*** 95% CI excludes zero ($p < 0.05$, two-sided).

Table 7. Continuation of Table 6

Year		(5) Excl. Right to Dev.	(6) Excl. Cultural Div.	(7) Excl. Israel/Palestine	(8) Excl. All
2000	NDE	-0.238*** [-0.380, -0.096]	-0.251*** [-0.387, -0.116]	-0.284*** [-0.474, -0.095]	-0.227*** [-0.440, -0.015]
	NIE	0.005 [-0.059, 0.068]	0.004 [-0.057, 0.065]	-0.015 [-0.103, 0.073]	-0.007 [-0.103, 0.090]
2005	NDE	-0.114*** [-0.227, -0.001]	-0.130*** [-0.242, -0.018]	-0.193*** [-0.342, -0.044]	-0.129 [-0.308, 0.050]
	NIE	-0.014 [-0.061, 0.032]	-0.018 [-0.065, 0.029]	-0.039 [-0.102, 0.025]	-0.024 [-0.095, 0.048]
2010	NDE	-0.071 [-0.191, 0.049]	-0.064 [-0.182, 0.053]	-0.088 [-0.238, 0.062]	-0.036 [-0.231, 0.158]
	NIE	0.048 [-0.003, 0.098]	0.039 [-0.010, 0.088]	0.083*** [0.018, 0.147]	0.070 [-0.012, 0.151]
2015	NDE	-0.051 [-0.190, 0.087]	-0.066 [-0.198, 0.067]	-0.104 [-0.266, 0.058]	-0.038 [-0.248, 0.172]
	NIE	0.091*** [0.025, 0.156]	0.096*** [0.033, 0.158]	0.132*** [0.055, 0.210]	0.099*** [0.001, 0.197]
2018	NDE	-0.021 [-0.177, 0.134]	0.007 [-0.153, 0.167]	-0.051 [-0.221, 0.119]	0.023 [-0.185, 0.231]
	NIE	0.104*** [0.027, 0.181]	0.094*** [0.016, 0.172]	0.142*** [0.056, 0.228]	0.140*** [0.038, 0.243]

*** 95% CI excludes zero ($p < 0.05$, two-sided).

Appendix 3. Definitions and Assumptions from Hayes et al. 2025

Definition 4. Let Z be a mean-zero random variable with cumulant generating function $\psi_Z(t) = \log \mathbb{E}[e^{tZ}]$.

1. Z is ν -sub-Gaussian for $\nu > 0$ if $\psi_Z(t) \leq t^2 \nu / 2$ for all $t \in \mathbb{R}$.
2. Z is (ν, b) -sub-gamma for $\nu, b \geq 0$ if $\psi_Z(t) \leq \frac{t^2 \nu}{2(1-bt)}$ and $\psi_{-Z}(t) \leq \frac{t^2 \nu}{2(1-bt)}$ for all $t < 1/b$.

Definition 5 (Regression Point Estimators). Define $\hat{D} = \begin{bmatrix} W & \hat{X} \end{bmatrix} \in \mathbb{R}^{n \times (2+p+k)}$. We estimate β_w and β_x via ordinary least squares as follows

$$\begin{bmatrix} \hat{\beta}_w \\ \hat{\beta}_x \end{bmatrix} = (\hat{D}^T \hat{D})^{-1} \hat{D}^T Y.$$

Similarly, we estimate Θ via ordinary least squares as

$$\hat{\Theta} = (W^T W)^{-1} W^T \hat{X}.$$

Definition 6 (Regression Variance Estimators). With notation as above, we define covariance estimators

$$\begin{aligned} \hat{\Sigma}_\beta &= \hat{A}_\beta^{-1} \cdot \hat{B}_\beta \cdot (\hat{A}_\beta^{-1})^T \quad \text{and} \\ \hat{\Sigma}_{\text{vec}(\Theta)} &= \hat{A}_{\text{vec}(\Theta)}^{-1} \cdot \hat{B}_{\text{vec}(\Theta)} \cdot (\hat{A}_{\text{vec}(\Theta)}^{-1})^T, \end{aligned}$$

where I_k is a $k \times k$ identity matrix, and letting $\hat{\xi}_{i\cdot} = \hat{X}_i - W_i \hat{\Theta}$, we define

$$\begin{aligned} \hat{A}_\beta &= \frac{\hat{D}^T \hat{D}}{n}, \quad \hat{B}_\beta = \frac{1}{n} \sum_{i=1}^n (Y_i - \hat{D}_i \hat{\beta})^2 \hat{D}_i^T \hat{D}_i, \\ \hat{A}_{\text{vec}(\Theta)} &= \frac{I_k \otimes W^T W}{n}, \quad \text{and} \quad \hat{B}_{\text{vec}(\Theta)} = \frac{1}{n} \sum_{i=1}^n \hat{\xi}_{i\cdot}^T \hat{\xi}_{i\cdot} \cdot \otimes W_i^T W_i. \end{aligned}$$

Assumption 4 (Regularity Conditions for Ordinary Least Squares M -estimation). We require standard regularity conditions for ordinary least squares estimates for both the outcome model and the mediator model. See Chapter 7 of Boos and Stefanski (2013) and Chapter 5 of Van der Vaart (2000) for additional discussion.

1. Define $\xi = X - \mathbb{E}[X | W] \in \mathbb{R}^{n \times k}$ to be the matrix of errors in the mediator regression. ξ_{ij} and $\xi_{i'j}$ are independent for $i \neq i'$ and all $j \in [k]$, and that $\mathbb{E}[\xi_{ij} | W] = 0$ for all $i \in [n], j \in [k]$. Further,

$$A_{\text{vec}(\Theta)} \equiv \mathbb{E} \left[\lim_{n \rightarrow \infty} \frac{1}{n} \sum_{i=1}^n W_i^T \cdot W_i \right] \in \mathbb{R}^{p \times p}$$

exists and is non-singular, and the matrix $B_{\text{vec}(\Theta)} \in \mathbb{R}^{k \times k}$ defined according to

$$\left[B_{\text{vec}(\Theta)} \right]_{j,j'} \equiv \mathbb{E} \left[\lim_{n \rightarrow \infty} \frac{1}{n} \sum_{i=1}^n \xi_{ij} W_i^T W_i \cdot \xi_{ij'} \right] \quad j, j' \in [k]$$

exists and has all entries finite.

2. Define $\widehat{D} = \begin{bmatrix} W & \widehat{X} \end{bmatrix} \in \mathbb{R}^{n \times (2+p+k)}$, let $\varepsilon = Y - \mathbb{E}[Y | D] \in \mathbb{R}^n$ be the vector of errors in the outcome regression. The ε_i are independent, and obey $\mathbb{E}[\varepsilon_i | D] = 0$ for all $i \in [n]$. Further, $A_\beta \equiv \mathbb{E} \left[\lim_{n \rightarrow \infty} \frac{1}{n} \sum_{i=1}^n D_i^T D_i \right] \in \mathbb{R}^{(p+k) \times (p+k)}$ exists and is non-singular, and

$$B_\beta \equiv \mathbb{E} \left[\lim_{n \rightarrow \infty} \frac{1}{n} \sum_{i=1}^n \varepsilon_i^2 D_i^T D_i \right]$$

exists and is finite.

Assumption 5 (Conditions for Two-stage Estimability). With notation as above, we assume that

1. The smallest non-zero eigenvalue λ_k of P grows at a sufficiently fast rate as a function of n relative to the noise parameters v_n and b_n . In particular,

$$(v_n + b_n^2) = o\left(\frac{\lambda_k^2}{n \log^2 n}\right)$$

2. The entries of the observation error vector ε have bounded second moments, i.e., $\max_{i \in [n]} \mathbb{E}[\varepsilon_i^2] < B$ for $B > 0$ not depending on n . Further, we assume that the error vector is such that $n^{-1} \sum_i \mathbb{E}[\varepsilon_i^4]$ is bounded as a function of n .

3. The rows $W_1, W_2, \dots, W_n$ of W are independent. Within each row $W_{i,\cdot}$, the elements may be dependent, but are marginally sub-Gaussian with fixed, shared parameter $\sigma > 0$.

4. The latent positions $X_1, X_2, \dots, X_n$ are such that $n^{-1} \sum_i \mathbb{E}[\|X_i\|^2]$ and $n^{-1} \sum_i \mathbb{E}[\|X_i\|^4]$ are bounded as functions of n .

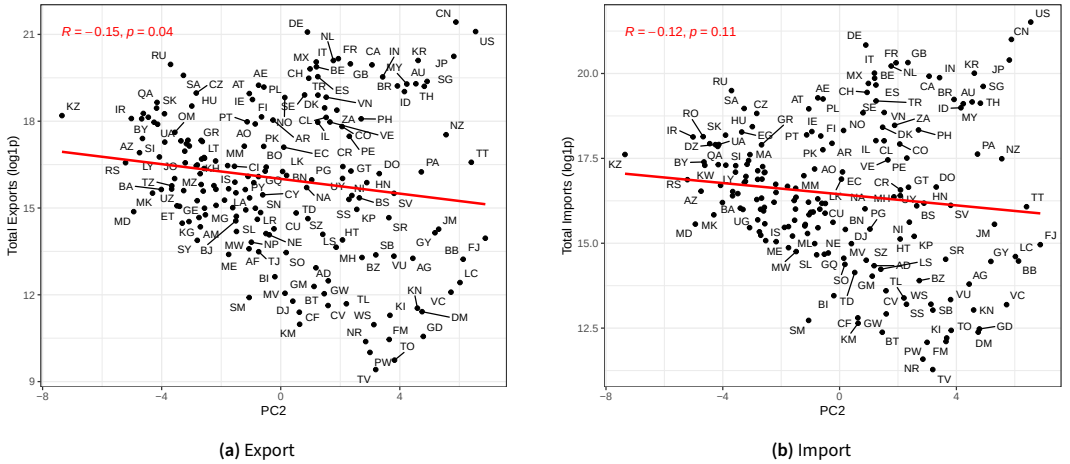


Figure 11. Correlation between the PC2 dimension and a country's export and import amounts. The x-axis shows the PC2 scores of each country, and the y-axis shows the log-transformed values of total export (11a) and import (11b) amounts. For illustration purposes, we only show the correlation between PC2 and trade volume in 2014, which aligns with our trade network plot (Figure 1).

Table 8. Regression of PC2 (sign-aligned) on trade volume and composition using 2014 trade network data. Models (1) and (2) use log-transformed import and export flows, respectively, as raw trade values are heavily right-skewed. Model (3) uses import share = imports/(imports + exports) untransformed, as it is already a bounded ratio ranging from 0 (pure exporter) to 1 (pure importer). Similar to Figure 4, PC2 scores are sign-normalized across years to ensure a consistent direction.

	Dependent variable: PC2 (sign-aligned)		
	(1)	(2)	(3)
Log imports	−0.171 (0.105)		
Log exports		−0.178** (0.086)	
Import share			3.067** (1.351)
Constant	2.807 (1.744)	2.848** (1.392)	−1.810** (0.827)
Observations	187	187	187
R ²	0.014	0.023	0.027
Adjusted R ²	0.009	0.017	0.022
Residual Std. Error (df = 185)	3.035	3.022	3.015
F Statistic (df = 1; 185)	2.631	4.292**	5.153**
<i>Note: Significance levels are indicated by stars: *p<0.1; **p<0.05; ***p<0.01</i>			